NENA Standard for Emergency Incident Data Object (EIDO)

Abstract: As agencies and regions move forward with implementing NG9-1-1 and IP based emergency communications systems, it is critical that they adhere to a standardized, industry neutral format for exchanging emergency Incident information between disparate manufacturer's systems located within one or more public safety agencies, and with other Incident stakeholders.

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The CRM will handle providing the remainder of the information needed on this page.
NENA Standard for Emergency Incident Data Object (EIDO)

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1 Executive Overview

There are many Functional Elements (FEs) within an NG9-1-1 system that are used to process emergency calls. Some of these FEs may be within a specific Agency, in another Agency, or elsewhere in an Emergency Services IP Network (ESInet). In many cases, an emergency call is related to, or results in the creation of an "Incident" as defined in the *NENA i3 Standard for Next Generation 9-1-1*, NENA STA-010 [4]. As public safety communication center personnel process emergency calls for service and their associated Incidents, new information about the Incident is obtained. There are many sources available to communication center personnel for obtaining new Incident information during call handling, Incident creation, dispatch, Incident monitoring, and post Incident analysis processes. Newly gathered information, as well as changes in Incident status, often needs to be passed on to other FEs, other involved agencies, and frequently to non-emergency entities authorized to receive emergency Incident information. As agencies and regions move forward with implementing NG9-1-1 and IP-based emergency communications systems, it is critical that they adhere to a standardized, industry-neutral format for exchanging emergency Incident information between disparate manufacturers' systems located within one or more public safety agencies, and with other Incident stakeholders.

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NENA
STANDARD DOCUMENT
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2 Document Conventions

NENA: The 9-1-1 Association improves 9-1-1 through research, standards development, training, education, outreach, and advocacy. Our vision is a public made safer and more secure through universally-available state-of-the-art 9-1-1 systems and better-trained 9-1-1 professionals. Learn more at https://www.nena.org.

2.1 Document Terminology

This section defines keywords, as they should be interpreted in NENA documents. The form of emphasis (UPPER CASE) shall be consistent and exclusive throughout the document. Any of these words used in lower case and not emphasized do not have special significance beyond normal usage.

1. MUST, SHALL, REQUIRED: These terms mean that the definition is a normative (absolute) requirement of the specification.
2. MUST NOT: This phrase, or the phrase "SHALL NOT", means that the definition is an absolute prohibition of the specification.
3. SHOULD: This word, or the adjective "RECOMMENDED", means that there may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.
4. SHOULD NOT: This phrase, or the phrase "NOT RECOMMENDED" means that there may exist valid reasons in particular circumstances when the particular behavior is acceptable or even useful, but the full implications should be understood and the case carefully weighed before implementing any behavior described with this label.
5. MAY: This word, or the adjective "OPTIONAL", means that an item is truly optional. One vendor may choose to include the item because a particular marketplace requires it or because the vendor feels that it enhances the product while another vendor may omit the same item. An implementation which does not include a particular option "must" be prepared to interoperate with another implementation which does include the option, though perhaps with reduced functionality. In the same vein, an implementation which does include a particular option "must" be prepared to interoperate with another implementation which does not include the option (except, of course, for the feature the option provides).
6. These definitions are based on IETF RFC 2119 [2].

2.2 Describing Artifacts

There are Artifacts [1] in this document, as defined in NENA Rules for Use of Version Control Repositories, NENA-ADM-012 [14] Artifacts defined in NENA documents, such as Representational State Transfer (REST) interfaces and JavaScript Object Notation (JSON) objects, usually use OpenAPI [3] as their formal description. In some cases, XML is used [15]

In the description of the interface or object in NENA documents, a table may appear that describes the parameters to the interface and the data objects. These tables are informative; the Artifact file is the normative documentation of the interface and data objects unless otherwise specified in the document. If there is a discrepancy between the table and the Artifact file, the Artifact file controls unless otherwise specified in the document.

In the tables, there is a column named "Required." The following three (3) values may occur in this column:

- "Required" means the parameter or member MUST be present. It MUST appear in the "required" section of the Artifact file.
- "Optional" means that the parameter or member MAY be present. It MUST NOT appear in the "required" section of the Artifact file.
- "Conditional" means that the presence of the parameter or member is conditional. This value alerts the reader that a business rule is specified, which controls the presence of the parameter or member. It MUST NOT appear in the "required" section of the Artifact file. This is because notation formats like YAML do not provide a way to support a conditional business rule.

Business rules are contained in this document for associated Artifacts. These business rules MUST be followed in implementing the standard.

2.3 Intellectual Property Rights (IPR) and Antitrust Policy

NOTE—The user's attention is called to the possibility that compliance with this standard may require use of an invention covered by patent rights. By publication of this standard, NENA takes no position with respect to the validity of any such claim(s) or of any patent rights in connection therewith. If a patent holder has filed a statement of willingness to grant a license under these rights on reasonable and nondiscriminatory terms and conditions to applicants desiring to obtain such a license, then details may be obtained from NENA by contacting the Committee Resource Manager identified on NENA's website at https://www.nena.org/ipr.

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2.4 Reason for Issue/Reissue

NENA reserves the right to modify this document. Upon revision, the reason(s) will be provided in the table below.

Document Number	Approval Date	Reason For Issue/Reissue
NENA-STA-021.1-2021	April 19, 2021	Initial Document
NENA-STA-021.1a-2021	April 19, 2022	Publishing error corrections: • Section 2.5 (page 15): agentRoleRegistryText example values now match values from the "Agent Roles" registry. • Section 2.10 (page 29): answerDate description now says "date and time stamp" instead of "date". • Section 2.6 (page 17): agencyType adds that values are from IANA urn:emergency:service:responder registry and the subregistries, and examples are provided.
NENA-STA-021.1b-2021	December 17, 2024	Publishing error corrections: • Section 2: Added normative text specifying that EIDO only conveys Incident current state, not history. • Section 2.20: Changed the minimum for resourceAttributeRegistryText from 0 to 1 because it is a Required Data Element. • Section 2.8: Set the maximum for involvedAgencyReference Data Element in the Link Data Component to 1 to match the normative schema. • Section 2.20: Added the sipIdentities Data Element to the Call Data Component that allows multiple values. It replaces the sipIdentity header from previous revision.

Document Number	Approval Date	Reason For Issue/Reissue
		• Section 2.11: updatedCbn element maximum in Callback Data Component is now set to 1 to be compliant with the schema. • Section 2.19: additionalDataComponent element clarification to indicate Additional Data in the provided-by section of a PIDF-LO should not be copied into an Additional Data Data Component. • Section 2.11: deviceContactHeader element in Call Back Data Component is now optional to match the minimum number of occurrences.
NENA-STA-021.1.1-202Y	[Month DD, YYYY]	New minor version that contains multiple changes such as new optional Data Components. This version is backwards compatible with NENA-STA-021.1b-2021.

3 Data Associated with an Emergency Incident

All registries mentioned in this document reference IANA (https://www.iana.org/protocols) "emergency" registry unless specified otherwise.

An EIDO SHALL represent the state of the Incident, as known by the sender, at the time it was sent. Only the known current state of the Incident is included. Every EIDO is logged when sent or received. For the history about the Incident, entities will query the Logging Service.

3.1 Data Components

Section 3 Data Associated with an Emergency Incident identifies the Data Elements associated with an emergency Incident grouped into various Data Components. Each Data Component section includes an initial (heading) section and a table of Data Elements.

The initial (heading) section of each Data Component contains the following information blocks:

- Data Component—the name of the Data Component (e.g., Agent Data component, Incident Data Component), displayed as the numbered section name.
- Data Component Use—identifies whether the Data Component is required, optional, or conditional in EIDO instances
- Minimum—the minimum number of occurrences of the Data Component allowed in EIDO instances.
- Maximum—the maximum number of occurrences of the Data Component allowed in EIDO instances. An unlimited number of occurrences is represented by "*".
- Child Of—identifies the potential parents of which a Data Component can be a child. See the parent component to determine the relationship between the child and parent.
- Data Component Description—a general description of the purpose and contents of the Data Component

The above information blocks are followed by a table that identifies the Data Elements included in the Data Components. Note that entire Data Components are included as a complex Data Element within their parent Data Component. In this case, the description identifies it as a complex Data Element (Data Component) and defines the relationship between the two Data Components.

The following information is included for each Data Element:

- JSON Name—name of the Data Element
- Use—identifies whether the Data Elements are required, optional, or conditional. Data elements that are conditional describe the conditions when they are required and when they are optional. Required Data Elements can exist within optional Data

Components. Required Data Elements of an optional Data Component are only required if the Data Component is included in an EIDO instance. For example, not all EIDO instances will contain a Dispatch Data Component since sufficient information to dispatch emergency resources to the Incident is not yet available or assigned resource statuses have not changed. However, if an EIDO instance contains a Dispatch Data Component, that Data Component will always contain the "Incident Type–Common" Data Element.

- Min—the minimum number of occurrences of the Data Element that may be included in the Data Component
- Max—the maximum number of occurrences of the Data Element that may be included in the Data Component. An unlimited number of occurrences is represented by "*".
- Description—a general description of the Data Element. For complex Data Elements (Data Components), the description identifies the relationship between the two Data Components.

Section 7 References includes sources and references that can be used to obtain additional information about related NENA standards, NIEM [5], and global justice terminology and standards.

Section 4 IANA Actions identifies registries that define the domain of values that MUST be used for specific EIDO elements.

3.2 Common Prologue Data Component

Data Component Use: Required

Minimum: 1

Maximum: *

Child Of: This Data Component is inherited by most Data Components but is not a child of any.

Data Component Description: Each Data Component, except the JSON Reference Data Component and Alternate Value Data Component, inherits this common set of elements to provide context and additional information.

Table 3-1 Common Prologue Data Component

JSON Name	Use	Min	Max	Description
$id	Required	1	1	An identifier that uniquely labels a Data Component. For Data Components where there is an identifier such as a Call Identifier, Agency Identifier, or Agent

JSON Name	Use	Min	Max	Description
				Identifier, that identifier is used for this member. For Data Components that do not have such an identifier, the form of the identifier is a URN [6] formed by the prefix "urn:emergency:uid:eidodatacomponent:", a unique string containing alphanumeric characters and if the agency that created the data component is known, the ":" character followed by the Agency Identifier of that Agency. For example, "urn:emergency:uid:eidodatacomponent:a56e556d871:idx.psap.allegheny.pa.us" is a properly formatted EIDO Data Component $id. The identifier MUST be the same for all instances of this specific Data Component for all EIDOs for the Incident, and unique from all other Data Components in the EIDO.
lastUpdateTimeStamp	Required	1	1	Timestamp of the last time the Data Component was modified. Must be in the W3C datetime format as specified in NENA-STA-010 [4].
updatedByAgencyReference	Required	1	1	Reference to the last Agency that updated the Data Component.
updatedByAgentReference	Optional	0	1	Reference to the last Agent that updated the Data Component. Must be present if the Data Component was created or updated by an Agent.
alternateValuesComponent	Optional	0	*	Alternate Values Data Component. Alternate values for Data Elements in this Data Component.
externalInformationSourceReference	Optional	0	*	References to External Information Source Data Components that are links to external information sources not otherwise defined in the EIDO.

JSON Name	Use	Min	Max	Description
mediaReference	Optional	0	*	References to Media Data Components that link to external media.
reportingPersonReference	Optional	0	1	Reference to a Person Data Component. The person that provided the information in the data component e.g. Victim, witness, etc.

3.3 JSON Reference Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Agent, Agency, Merge, Link, Incident, Call Data, Dispatch, Notes, Additional Data, Emergency Resource, Alarms and Sensors

Data Component Description: This JSON Reference Data Component is used by other Data Components to reference Data Components present elsewhere in the EIDO document.

Table 3-2 JSON Reference Data Component

JSON Name	Use	Min	Max	Description
$ref	Required	1	1	Data element containing the identifier ($id) of the referenced Data Element.

3.4 Media Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Media Data Component is a URI link to media. This includes a link to a static file as well as streaming media.

Table 3-3 Media Data Component

JSON Name	Use	Min	Max	Description
uri	Required	1	1	A URI, which when dereferenced, will retrieve the media.
contentType	Required	1	1	An entry from the IANA "Media Types" registry (formerly known as MIME types) that indicates the type of media.
label	Optional	0	1	A UTF-8 label used to describe the media in a human readable format.

3.5 Emergency Incident Data Object Data Component

Data Component Use: Required

Minimum: 1

Maximum: 1

Child Of: None

Data Component Description: This is the parent Data Component that contains all the other Data Components to form an EIDO. This Data Component MUST always be present. A functional element receiving an EIDO for an Incident where the state is already known and needs to know what changed, it can compare the previously known state with the received EIDO.

In this Data Component, the identifier ($id) is the Incident Tracking Identifier. The Incident Tracking Identifier is an identifier assigned by the first element in the ESInet. The form of an Incident Tracking Identifier is defined in NENA-STA-010[1] [4].

Incident Tracking Identifiers are globally unique and are associated with a single emergency Incident. The Incident Tracking Identifier can be associated with one or more emergency calls. It is carried through to any Incident resulting from an emergency call. It may or may not be the same as the local Incident ID.

When EIDO is exchanged not as part of an Incident, it contains an Incident Tracking Identifier as identified in NENA-STA-010 [4] where "unique string" is "0". Example: urn:emergency:uid:incidentid:0:police.state.pa.us

The callbackComponent, agencyComponent, agentComponent, notesComponent, additionalDataComponent, locationComponent, personComponent, vehicleComponent,

[1] NENA-STA-010 includes text to handle the case of sending the EIDO by reference or value in a transferred SIP call.

cautionComponent, and mediaComponent MUST be linked to at least one other Data Component in the EIDO. An EIDO with these Data Components not being linked by other Data Components MUST NOT be rejected for backward compatibility with version 1 of this document. Additionally, when these Data Components are not linked by any other Data Component, they SHOULD be assumed to be associated with the Incident represented by the EIDO.

Table 3-4 Emergency Incident Data Object Data Component

JSON Name	Use	Min	Max	Description
eidoVersion	Required	1	1	Version of the EIDO structure. For this version of the standard the value MUST be "1.1". For backward-compatible changes to the structure, the minor version number will be incremented. For any non-backward-compatible changes to the structure, the major version number will be incremented.
eidoExtensionId	Optional	0	*	EIDO extension IDs identifying extensions that are used in the EIDO instance. Values are to be included in the EIDO-EidoExtensionId registry.
issuingElementIdentification	Required	1	1	An identifier in the format defined in NENA-STA-010 [4]. Identifies the element that created the EIDO.
mergeComponent	Optional	0	*	Complex Data Element (Data Component). Contains merge and split information related to the Incident.
linkComponent	Optional	0	*	Complex Data Element (Data Component). Contains link information related to the Incident.
incidentComponent	Optional	0	1	Complex Data Element (Data Component). Contains general information about the Incident.
callComponent	Optional	0	*	Complex Data Element (Data Component). Contains information about calls associated with the Incident.

JSON Name	Use	Min	Max	Description
callbackComponent	Optional	0	*	Complex Data Element (Data Component). Contains information about how to call a person.
dispatchComponent	Optional	0	*	Complex Data Element (Data Component). Contains dispatch information related to the Incident.
notesComponent	Optional	0	*	Complex Data Element (Data Component). Contains Incident notes and comments associated with the Incident.
emergencyResourceComponent	Optional	0	*	Complex Data Element (Data Component). Identifies emergency resources involved with the Incident.
alarmsSensorsComponent	Optional	0	*	Complex Data Element (Data Component). Identifies Alarms/Sensors associated with the Incident.
agencyComponent	Required	1	*	Complex Data Element (Data Component). Identifies all agencies involved with the Incident.
agentComponent	Conditional: Must be provided if an Agent is involved in the Incident	0	*	Complex Data Element (Data Component). Identifies all Agents involved with the Incident.
additionalDataComponent	Optional	0	*	Complex Data Element (Data Component). All Additional Data related to the Incident except the objects found in the <provided-by> field of a PIDF-LO.
locationComponent	Conditional: Required if a location is known about the Incident	0	*	Complex Data Element (Data Component). All locations related to the Incident.
personComponent	Optional	0	*	Complex Data Element (Data Component). Every person related to the Incident.

JSON Name	Use	Min	Max	Description
vehicleComponent	Optional	0	*	Complex Data Element (Data Component). Every vehicle related to the Incident.
cautionComponent	Optional	0	*	Complex Data Element (Data Component). Caution information related to the Incident.
mediaComponent	Optional	0	*	Complex Data Element (Data Component). Media information related to the Incident.
sharedCommunicationResourceComponent	Optional	0	*	Complex Data Element (Data Component). All shared resources used for shared communication between Agents and agencies for this Incident.
externalInformationSourceComponent	Optional	0	*	All external information sources not otherwise defined in the EIDO.
itemComponent	Optional	0	*	Complex Data Element (Data Component). All item information referenced in this document are listed here.
organizationComponent	Optional	0	*	Complex Data Element (Data Component). All organization information referenced in this document is listed here.
unitEmergencyStateComponent	Optional	0	*	Complex Data Element (Data Component). All unit emergency state information referenced in this document are listed here.

3.6 Agent Data Component

Data Component Use: Conditional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component contains information about Agents (e.g., call takers, dispatchers, supervisors, responders) that are involved in the Incident. There may be multiple Agent Data Components in cases where both a call taker and dispatcher are involved in an Incident, where multiple dispatch agencies are associated with the same Incident, and similar situations.

Rarely, as in the case of automatically dispatched responses, the Agent may be an automaton (automated system) such as an Interactive Media Response (IMR). Automatons that are actively involved in an Incident or call MUST be assigned an Agent ID that follows the i3 naming conventions (see NENA-STA-010 [4] for more information). In this Data Component, the identifier ($id) is the Agent Identifier.

Table 3-5 Agent Data Component

JSON Name	Use	Min	Max	Description
agentJcard	Optional	0	1	The Agent's contact information.
agentWorkstationPositionIdentification	Conditional: Required if the Agent's workstation has a position identification	0	1	The workstation position ID within the Agent's Agency.
agentRoleRegistryText	Required	1	*	The role of the Agent in the context of the Incident—Dispatching, Call Taking, etc. The acceptable roles are defined in the IANA "Agent Roles" registry in NENA-STA-010 [4]. Some roles are flagged as modifiers. Modifiers should be added immediately after the Role they apply to.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the Agent.
agencyReference	Required	1	1	Reference to an Agency Data Component. Identifies the Agency employing or contracting with the Agent.
agentCallLegState	Optional	0	*	Call Leg State Data Component. Used to represent the Agent call leg state when it is different from the global call state. SHOULD be omitted if the state of the call leg is the same as the call.

3.7 Call Leg State Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Agent, Agency, Person

Data Component Description: This Data Component is used to represent the call leg state when it is different from the global call state. This Data Component SHOULD be omitted if the state of the call leg is the same as the call.

Table 3-6 Call Leg State Data Component

JSON Name	Use	Min	Max	Description
callReference	Required	1	1	Reference to a Call Data Component that represents the call the call leg is related to.
callStateRegistryText	Required	1	1	State of the call leg.

3.8 Agency Data Component

Data Component Use: Required

Minimum: 1

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: Every EIDO will include at least one instance of this Data Component in the Agency Data Component of the Emergency Incident Data Object in order to identify the Agency creating the EIDO.

Many Incidents have one owner, a specific Agency. Sometimes, ownership changes from one owner to another. In some jurisdictions, there can be more than one owner. Normally, ownership is passed from the current owner to another, but there are circumstances where ownership is unclear, and ownership needs to be claimed. The Agency Data Component provides a mechanism for establishing the Agency that owns the Incident associated with the Incident Tracking Identifier contained in the Incident Data Component or for removing current ownership from that Incident.

In this Data Component, the identifier ($id) is the Agency Identifier. The Agency Identifier identifies the Agency. See NENA-STA-010 [4] Agency Identifier section for the format and requirements of the identifier Data Element within the identity component.

The Agency whose identifier is found within this component MUST have a record in the Service/Agency Locator as defined in NENA-STA-010 [4].

Table 3-7 Agency Data Component

JSON Name	Use	Min	Max	Description
agencyJcard	Optional	0	1	The Agency's contact information.
agencyRoleDescriptionRegistryText	Required	1	*	The role of the Agency in relation to the Incident. Valid roles are available in the IANA "EIDO-AgencyRole" registry and include: Dispatching, Dispatched, CallReceiving, and TransferredTo.
agencyType	Required	1	*	One or more members of a list of available provider and Agency types. The Agency types are taken from the IANA "urn:emergency:service:responder" registry and the subregistries defined for that registry. Values exclude the "urn:emergency:service:responder." prefix. Example values: "police.federal.fbi", "fire.forest", "poison_control"
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text about the Agency in the context of the Incident.
incidentOwningAgencyIndicator	Optional	0	1	Boolean Data Element that, if true, indicates that the Agency associated with the Agency ID contained in this Data Component owns, or if false, does not own the Incident associated with the Incident Tracking Identifier in the EIDO Data Component. Once set to true, it should only be set to false by the Agency that originally set it true. Used to enable the exchange and update of Incident ownership information. Only one Agency Data Component can have this set to true in the EIDO.

JSON Name	Use	Min	Max	Description
actingOnBehalfOfAgencyReference	Optional	0	*	Reference to an Agency Data Component. If the Agency is operating on behalf of another Agency as its backup, this contains a reference to the other Agency's Agency Data Component.
agencyCallLegState	Optional	0	*	Call Leg State Data Component used to represent the Agency call leg state when it is different from the global call state. SHOULD be omitted if the state of the call leg is the same as the call or when the call is known to be connected to an Agent. When the call is connected to an Agent, the Agent's Agency call leg state SHOULD be omitted and assumed to be the same as the Agent call leg state.

3.9 Merge Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This optional Data Component is used to indicate the existence of a merged Incident Tracking Identifier or to split an Incident. The presence of a Merge Data Component indicates that another Incident Tracking Identifier has been merged with or is being split from the Incident Tracking Identifier contained in the EIDO Data Component.

When merging two Incidents, two EIDO will be created. One EIDO will have the surviving Incident ID in the EIDO Data Component, the retired Incident ID in the incidentTrackingIdentifier Data Element, and the incidentMergeDirectionCode set to REPLACED. The other EIDO will have the retired Incident ID in the EIDO Data Component, the surviving Incident ID in the incidentTrackingIdentifier Data Element, and the incidentMergeDirectionCode set to REPLACING.

NENA-STA-010 [4] allows Incidents to be unmerged, canceling a merge operation. When this happens, EIDOs SHOULD be sent for both Incidents without the Merge Data

Component that was added for the merge. A more explicit method of representing the unmerge will be defined in a future version of this document.

Table 3-8 Merge Data Component

JSON Name	Use	Min	Max	Description
incidentTrackingIdentifier	Required	1	1	The Incident Tracking Identification of the Incident that is being merged with, or split from, the Incident represented by the Incident Tracking Identifier contained in the EIDO Data Component. See incidentMergeDirectionCode to determine the direction of the split/merge.
incidentMergeDirectionCode	Required	1	1	The role of the IncidentTrackingIdentifier in this Data Component. Indicator is: "REPLACED"—The Incident Tracking Identifier in this Data Component contains the retired Incident Tracking Identifier. Only applies to a merge operation. "REPLACING"—The Incident Tracking Identifier in this Data Component contains the surviving Incident Tracking Identifier. Only applies to a merge operation. "SPLIT"—The Incident Tracking Identifier contained in this Data Component is split from the Incident Tracking Identifier contained in the Incident Data Component. The other Data Components contained in the EIDO contain the Data Elements of the split Incident.

3.10 Link Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This optional Data Component is used to indicate the existence of linked calls and Incidents. A Link Data Component indicates that an Incident has been linked to the Incident Tracking Identifier in the EIDO Data Component. Incidents are linked when it is determined that while they are separate Incidents, they are related in some way. When a link is declared, both Incident Tracking Identifiers continue to be used to track the individual Incidents.

Incidents may be linked in a hierarchical relationship. For more information on hierarchical Incidents, see the Incident Tracking Identifier section of NENA-STA-010 [4].

Table 3-9 Link Data Component

JSON Name	Use	Min	Max	Description
incidentTrackingIdentifier	Required	1	1	The Incident Tracking Identifier of the Incident that is being linked to the Incident represented by the Incident Tracking Identifier contained in the EIDO Data Component. The nature of the link is defined by the linkDirectionCode below.
transactionReasonText	Optional	0	1	Free format narrative description of the reason for the link.
involvedAgencyReference	Optional	0	1	Other Agency involved with the linked Incident, if known.
linkDirectionCode	Required	1	1	The direction of the link. If the value of the Link Indicator is: “PARENT”—The Incident Tracking Identifier contained in this Data Component is the parent of the Incident Tracking Identifier contained in the EIDO Data Component. “CHILD”—The Incident Tracking Identifier contained in this Data Component is the child of the Incident Tracking Identifier in the EIDO Data Component.

JSON Name	Use	Min	Max	Description
				"RELATED"—The Incident Tracking Identifier contained in this Data Component is related to the Incident Tracking Identifier in the EIDO Data Component, without any parent-child relationship. "UNLINK"—The Incident Tracking Identifier contained in this Data Component is unlinked from the Incident Tracking Identifier contained in the Incident Data Component.

3.11 Incident Data Component

Data Component Use: Optional

Minimum: 0

Maximum: 1

Child Of: Emergency Incident Data Object

Data Component Description: The Incident Data Component is optional and is used to exchange general information about emergency Incidents gathered by emergency Agents, emergency responders, reporting parties (callers), and devices reporting emergency Incidents. Agencies responding to the same Incident may have different versions of the Incident Data Components for the Incident. Each Agency's version of the Incident Data Component contains information that is relevant to their Agency but may also contain shared information that is common to both agencies.

This module is used to exchange Incident update information, as well as for exchanging initial Incident creation information. For example, in high priority Incidents, only partial information may be exchanged between call takers and dispatchers (i.e., the Incident's type and location) while additional information is being collected. This Data Component is used for the initial, high priority exchange and the subsequent exchange containing the additional information collected after the initial exchange was completed. When multiple callers report a single Incident, this Data Component is used to update involved Agents and responders about new information gathered from the other callers.

The Incident Data Component is also used to exchange general Incident information developed during dispatch operations. Call takers, dispatchers, emergency resources, and emergency devices can enter information exchanged/carried by this Data Component.

Table 3-10 Incident Data Component

JSON Name	Use	Min	Max	Description
incidentTypeInternalCode	Optional	0	1	An alphanumeric code indicating the category of the Incident. This is the internal code used by the local agencies involved in the Incident.
incidentTypeInternalText	Optional	0	1	Textual description indicating the category of the Incident. This is the internal text used by the local agencies involved in the Incident.
incidentTypeCommonRegistryText	Required	1	1	Incident type code that is available in the IANA "EIDO-IncidentType-Common" registry and that most closely corresponds to the Incident Type internal code. APCO has developed an ANS set of globally unique common Incident type codes in *Public Safety Communications Common Incident Types for Data Exchange*, APCO 2.103 [7], which form the basis for this registry. Each Agency should maintain a mapping of its Internal Incident Types (IncidentTypeInternal) to the list of Common Incident Types (IncidentTypeCommon). The Common Incident Type should be selected from this mapping when the EIDO is created to identify the Incident type using a common code that is globally understood.
incidentStatusInternalText	Optional	0	*	An alphanumeric code indicating the status of the Incident (active, closed, structure cleared, etc.). This is the internal code used by the local agencies involved in the Incident.
incidentStatusCommonRegistryTe xt	Optional	0	*	Incident status code that is available in the IANA "EIDO-Incident Status" registry and that most closely corresponds to the Incident Status-Internal. Typically used to track significant changes in an Incident's

JSON Name	Use	Min	Max	Description
				status. Each Agency should maintain a mapping of its internal Incident status (IncidentStatusInternal) to the list of common Incident status (IncidentStatusCommon). The common Incident status should be selected from this mapping when an EIDO is created to identify the Incident status using a common code that is globally understood.
internalIncidentId	Optional	0	1	The Internal Incident ID as an alphanumeric string assigned by the Agency involved in the Incident. Used to exchange Incident information between systems using the same internal Incident IDs. Maintained for conformance with legacy systems.
agentReference	Optional	0	*	Reference to Agent Data Components. Identifies the Agents (could be either an Agent in a communication center or an emergency responder) that are involved in this Incident.
locationReference	Optional: Required if available	0	*	Reference to a Location Data Component. Contains Incident location information entered or updated by an Agent receiving a call associated with the Incident.
personReference	Optional	0	*	Reference to a Person Data Component. Contains person information entered or updated by an Agent receiving a call associated with the Incident.
organizationReference	Optional	0	*	Reference to an Organization Data Component. Contains information about organizations involved with the Incident.
vehicleReference	Optional	0	*	Reference to a Vehicle Data Component. Contains vehicle information entered or updated by

JSON Name	Use	Min	Max	Description
				an Agent receiving a call associated with the Incident.
documentIdentification	Optional	0	*	The Document Identification connects the Incident to one or more associated follow-up reports and investigations. Each responding Agency may have its own Document Identification. Also used by the Agency supervisor and other personnel to track the status of reports.
reportNumberType	Conditional: MUST be populated if documentIdentification is present	0	1	This element will be removed in a future version of this document. Report number type codes that are available in the IANA "EIDO-ReportNumberType" registry ; it may be New or Reopened. *This is maintained for backward compatibility.*
incidentPriorityInternalText	Optional	0	1	Priority of the Incident as alphanumeric text for the Agency being dispatched. This value may only be meaningful to the Agency providing the information and other closely cooperating agencies. Note that different responding agencies may assign different priorities to the same Incident; for example, a high priority fire Incident may be a medium priority law enforcement Incident.
incidentCommonPriorityNumber	Optional	0	1	Globally understood numeric Incident priority that ranges from 0 to 10, with 10 being the highest priority and 0 being the lowest priority. The Local Priority, described above, should be mapped to this (Common Priority) Data Element so that all involved and interested agencies can determine the relative priority of the Incident.

JSON Name	Use	Min	Max	Description
beatOrDispatchGroupText	Optional	0	1	The beat or dispatch group that contains the Incident. Note that each Agency involved in the Incident may have its own beat or dispatch group.
dispositionComponent	Optional	0	*	Complex Data Element. Contains Incident disposition information entered or updated by a dispatch Agent and/or an emergency responder.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the Incident.

3.12 Shared Communication Resource Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is optional and is used to exchange information about shared resources used for shared communication between Agents and agencies for this Incident. This would include information about radio channels, talk groups, chat rooms, etc.

Each Shared Communication Resource type has a schema specified in an associated registry entry. See Section 4.2 "EIDO Shared Communication Resource Type" Registry.

Table 3-11 Shared Communication Resource Data Component

JSON Name	Use	Min	Max	Description
sharedCommunicationResourceType	Required	1	1	Type of shared communication resource (e.g., radioChannel, talkGroup). Values limited to those in the IANA "EIDO-SharedCommunicationResourceType" registry.
sharedCommunicationResourceByReferenceUrl	Conditional: MUST be populated if	0	1	The URL is a link to a Shared Communication Resource Data Component. This allows the

JSON Name	Use	Min	Max	Description
	sharedCommunicationResourceByValue is not populated. Otherwise, it MUST be omitted.			dereferencing of the shared communication resource detail associated with an Incident.
sharedCommunicationResourceByValue	Conditional: MUST only be populated if sharedCommunicationResourceByReferenceUrl is not populated. Otherwise, it MUST be omitted.	0	1	Shared Communication Resource Data Component provided by value.
connectedSharedCommunicationResources	Optional	0	*	Reference to Shared Communication Resource Data Components. Represents the other communication resources that share a communication path with this one. The referenced components should be part of the EIDO and reference this instance, creating a circular reference. Implementations MUST take this fact into consideration to avoid the circular reference exception.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains notes and comments related to the Additional Data that were entered by Agents and emergency responders.

3.13 Call Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: The Call Data Component is optional and is used to exchange call information about the Incident received and collected by the Agent identified in this Data Component. There can be more than one call about an Incident, and thus more than one instance of this Data Component can be in an EIDO. Some of the information in this Data Component is contained in the call, some is additional data associated with the call, and other information is collected by the Agent.

For non-interactive calls, as described by RFC 8876 [8], a call Data Component will contain the additional data that came in with the SIP MESSAGE, including the Common Alerting Protocol (CAP) message. A call component describing a non-interactive call MUST have the standardSecondaryCallType set to "nonInteractive" and callStateRegistryText set to "callEnd". Since the call Data Component is not populated in EIDO for calls that are not ongoing, additional EIDO should include an Incident Data Component referencing the appropriate Data Component (e.g., person, vehicle, or location, which are referencing the CAP message's additional data block).

This Data Component can represent either an inbound call to an Agency or outbound from an Agency (e.g., Callback). In the case of an emergency call and of an emergency Callback or Follow-Up Call, the Callback Data Component represents the contact information of the person who originated the original emergency call. In a case of a call that is not an emergency Callback or Follow-Up Call between parties associated with an Incident, the Callback Data Component is to be omitted, and Agent, Agency, or person references MUST be used to represent parties in the call.

In this Data Component, the identifier ($id) is the Call identifier. The Call identifier is automatically created by the first ESRP in the first ESInet that handles a call. Call Identifiers are globally unique and are only valid for a specific call.

Table 3-12 Call Data Component

JSON Name	Use	Min	Max	Description
relatedCallIdentifier	Optional	0	*	Call Identifier of calls that are related to this call (e.g., Callbacks).
queueIdentifier	Optional	0	1	An identifier of a queue where the call is currently on.
standardPrimaryCallType	Required	1	1	Primary call type. Values limited to those in the IANA "LogEvent CallTypes" registry with "Primary" classification.
standardSecondaryCallType	Optional	0	1	Secondary call type. Values limited to those in the IANA "LogEvent

JSON Name	Use	Min	Max	Description
				CallTypes" registry with "Secondary" classification.
localCallType	Optional	0	1	Locally defined call type.
direction	Required	1	1	The "direction" member has one of these values, "incoming", "outgoing" or "internal", where "incoming" means received by the EIDO sender, "outgoing" means sent by the EIDO sender, and "internal" means both caller and recipient are within the EIDO sender.
priorityIdentifier	Optional	0	1	Call Priority Identifier defined in *NENA NG9-1-1 PSAP and ECC Specifications for the NENA i3 Solution*, NENA-STA-023 [9]. It is used to describe the priority of a call; if present, it has one of these values: "highest", "high", "normal", "low", "lowest".
additionalDataReference	Optional	0	*	Reference to an Additional Data Data Component. Additional information about a call received that is involved in or related to the Incident. There may be multiple data providers for one call. Additional Information is defined in RFC 7852 [10]
callStartTimestamp	Required	1	1	Date and time stamp of when the call was received. It SHALL be represented by the "date-time" datatype described in RFC 3339, Section 5.6, as specified in NENA-STA-010 [4].
answerDate	Optional	0	1	Date and time stamp on which the call was answered by an Agent.
callStateRegistryText	Required	1	1	Current call status (when the EIDO was created) from the available call statuses in the IANA "Call States" registry (e.g., callBegin, callAnswered, callEnd, partyAdd).
agentReference	Conditional:	0	*	Reference to an Agent Data Component. Identifies the Agent(s)

JSON Name	Use	Min	Max	Description
	Must always be present if one or more Agents are involved in the call			that is currently directly involved with the call described in this Data Element.
callBackReference	Conditional: Required if available	0	1	Reference to a Call Back Data Component. Identifies how the original emergency caller described in this Data Component can be called back.
verstat	Conditional: Required if known	0	1	Value of the verstat parameter. Will be removed in a future revision, being replaced by verstat Data Element in the Callback Data Component.
sipIdentity	Conditional: Required if known	0	1	Content of the first SIP identity header. Required if known. Will be removed in a future revision, being replaced by sipIdentities Data Element in the Callback Data Component.
sipIdentities	Conditional: Required if known	0	*	Content of the SIP identity headers associated with the caller identity. Will be removed in a future revision, being replaced by sipIdentities Data Element in the Callback Data Component.
locationReference	Conditional: Must be present if location is presented with the call	0	*	Reference to a Location Data Component. Contains call location information received with the call or updated by the Agent receiving the call.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the call.
personReference	Conditional: Must be present if the	0	*	Reference to a Person Data Component. Contains information about the parties on the calls that

JSON Name	Use	Min	Max	Description
	call is a SIP session			are not represented by the agentReference or agencyReference.
organizationReference	Optional	0	*	Reference to an Organization Data Component. Contains information about organizations involved with the call.
agencyReference	Optional	0	*	Reference to an Agency Data Component. Contains information about agencies connected to the call. Used to know which agencies are connected to the call when information about the connected Agent within these agencies are not known. This Data Element will be made conditional in a future version of this document.

3.14 Callback Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is used to exchange information about how to contact the Incident's reporting parties. When the original emergency call is a SIP call, deviceContactHeader MUST be included. updatedCbnReference is introduced in this version to eventually replace updatedCbn. An implementation from this version MUST populate both Data Elements. updatedCbn will be deprecated in a future revision of this document.

Table 3-13 Callback Data Component

JSON Name	Use	Min	Max	Description
callBackInformationUri	Conditional: Required if available	0	*	Information that enables Agents and responders to Callback. URI(s) found in the P-Asserted-Identity header, if provided, else the URI in the From header.
deviceContactHeader	Conditional: Required if the call	0	1	Content of the original emergency call's SIP INVITE contact header. Information that enables Agents and

JSON Name	Use	Min	Max	Description
	includes a contact header.			responders to possibly reach (Callback) the device that initiated the call. Note that this information is only guaranteed to be valid during the call and for a few minutes after it ends.
verstat	Conditional: Required if known	0	1	Value of the verstat parameter.
sipIdentities	Conditional: Required if known	0	*	Content of the SIP identity headers associated with the caller identity.
updatedCbn	Optional	0	1	Complex Data Element. Identifies a telephone number or SIP equivalents that can be used to contact the individual who made the call described in this Data Component.
updatedCbnReference	Optional	0	*	Reference to a Updated Callback Number Data Component. Identifies additional telephone numbers and SIP equivalents that can be used to contact the individual who made the call described in this Data Component.

3.15 Updated Callback Number Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Callback

Data Component Description: This Data Component is optional and is used to exchange information about additional phone numbers that can be used to contact the Incident’s reporting parties.

Table 3-14 Updated Callback Number Data Component

JSON Name	Use	Min	Max	Description
updatedCbnIdentifierUri	Required	1	1	This Data Element (in the form of a URI) is used to track additional

JSON Name	Use	Min	Max	Description
				telephone numbers or SIP equivalents that can be used to contact the reporting party of the parent call.
updatedCbnCallerDescription	Optional	0	1	Descriptive (alphanumeric) text that provides additional information about the updated Callback number such as hours to use it, days to use it, and the type of number (e.g., work, home, friend).

3.16 Dispatch Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component contains dispatch related information. It allows updates to be sent and received between Incident Handling FEs and Dispatch FEs, between different Dispatch FEs that are working the same Incident, and enables exchanging information provided directly by emergency responders. It can also be used to provide dispatch-related status updates to involved agencies and authorized stakeholders.

Table 3-15 Dispatch Data Component

JSON Name	Use	Min	Max	Description
agencyReference	Optional	0	1	Reference to an Agency Data Component. Agency that was dispatched through the action performed in this Data Component. Note that if several agencies are dispatched (one fire and one police, two fire agencies, etc.), there MUST be a separate instance of the Dispatch Data Component for each dispatched Agency. Agencies are globally unique.
agentReference	Optional	0	1	Reference to an Agent Data Component. Identifies the Agent that

JSON Name	Use	Min	Max	Description
				completed the dispatch operation described in, and/or entered the information contained in this Data Component. Should be provided if the information is available.
emergencyResourceReference	Optional	0	*	Reference to an Emergency Resource Data Component. Contains information about emergency responders assigned (dispatched) to the Incident, as well as their status and location updates.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the dispatch.

3.17 Disposition Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Incident Information

Data Component Description: This Data Component contains Agency specific and standardized disposition codes assigned to an Incident. Multiple disposition codes per Incident are supported. Either a responder or a dispatcher can close an Incident and assign a final disposition code to it. DispositionDescriptionText is required when dispositionCategoryText is present. Otherwise, it is optional.

Table 3-16 Disposition Data Component

JSON Name	Use	Min	Max	Description
dispositionCommonRegistryCode	Required	1	1	An Agency assigns a disposition to an Incident when its participation in the Incident ends. The disposition code indicates whether follow-up reports are required and other information about the Incident such as whether it resulted from a false or actual alarm. They are used to exchange the status and follow-up

JSON Name	Use	Min	Max	Description
				requirements of an Incident upon its closure. The disposition codes are drawn from the IANA "EIDO-CommonDispositionCode" registry containing common disposition codes for Police, Fire EMS disciplines.
dispositionPrimaryIndicator	Optional	0	1	A designation of whether the common disposition code is the primary disposition code for the Incident. Note that multiple primary codes are allowed, but some systems may not be able to handle more than one primary common disposition code. It is possible that no codes are marked as primary. The value is Boolean, where True is a primary disposition code.
dispositionCategoryText	Optional	0	1	An Agency-specific, alphanumeric code that indicates how the Incident was closed. The Common Disposition Code, referenced above, should be mapped to the closest value of this Data Element.
dispositionDescriptionText	Optional	0	1	Descriptive text describing the Disposition Code Internal. Disposition codes may be Agency specific, and this field explains the meaning of the internal disposition code.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the disposition code.

3.18 Notes Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Description: This Data Component is typically populated by emergency service Agents and responders and occasionally by automated devices. There may be multiple notes from the same Agent, and there may be notes from multiple Agents and agencies.

It is difficult to determine when a note should be considered history, and thus not sent in an EIDO. For that reason, if a Note is referenced by a Data Component sent in the EIDO, then the note SHALL be included in the EIDO and remains part of what is considered the current state. If a system permits notes to be deleted, then deleted notes SHALL NOT be sent in subsequent EIDOs.

Table 3-17 Notes Data Component

JSON Name	Use	Min	Max	Description
notesActionComments	Required	1	1	Notes created by an Agent entered in HTML fragments, as supported by NIEM, and shall be limited to 16 MB. HTML is used to allow multimedia data to be contained in the notes. Security issues may arise from embedding scripts, images, and other references, including JavaScript, in notes, and the receiving system may ignore or filter out such embedded information.
agentReference	Optional	0	1	Reference to an Agent Data Component. Identifies the Agent and Agency that entered the note contained in this Data Component. Defaults to the Agency Information included in the EIDO Data Component, if this Data Component is not present.
cautionReference	Optional	0	*	Reference to a Caution Data Component. Used to raise special attention to circumstances related to this Note that might threaten the safety of people or property.

3.19 External Information Source Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is used to convey a link to an external information source not otherwise defined in the EIDO. The referenced external information source is likely to apply its own data rights management policies while responding to any agencies trying to access the external data. This is not a replacement for Additional Data, but can be used to provide a reference to an Identity Searchable Additional Data Repository (IS-ADR). A data provider SHOULD use the mechanism described in RFC 7852 [10] for any new sources of data about a call, person, or location.

Examples of these external sources would be IS-ADR, Criminal Justice Information Services (CJIS), Canadian Police Information Centre (CPIC), National Crime Information Center (NCIC), and Departments of Motor Vehicles (DMV).

Table 3-18 External Information Source Data Component

JSON Name	Use	Min	Max	Description
externalSourceRegistryText	Required	1	1	An identifier that SHOULD come from the external information source registry. The registry values are linked to a description in the registry that gives details about how to access the data.
externalSourceAccessString	Optional	0	1	A string value that can be used to query the data source. For example, a URL or a query string.

3.20 Person Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is used to exchange information about people associated with an Incident, including callers, suspects, victims, witnesses, and other individuals involved in the Incident. The information is provided by reporting parties and emergency responders. Responders can enter the information either directly through their mobile data computers or via their assigned dispatchers.

An instance of Person Data Component MUST be created along with a Call Data Component at the beginning of an emergency call. The Person Data Component MUST have the personIncidentRoleRegistryText element set to "caller" and the callBackReference element refers to the same Callback Data Component designated by the Call Data Component. This business rule MUST be followed even in cases when it is unknown if the Callback information will allow an Agency to contact the person who initiated the emergency call. If the emergency call included a SubscriberInfo Additional Data block, the Additional Data Data Component reference for that block and its associated ProviderInfo block SHOULD be specified in an additionalDataReference element. At the end of the emergency call, while the EIDO may no longer include a Call Data Component, the caller's details will continue to be found in the Person Data Component.

When the EIDO is used to describe a Callback Call, a Person Data Component linked by the Callback's Call Data Component SHOULD contain one or more references to Location Data Components where the Location Data Components MUST have the "locationTypeDescriptionRegistryText" set to "Caller", populated with the location(s) provided with the incoming call.

Table 3-19 Person Data Component

JSON Name	Use	Min	Max	Description
personIncidentRoleRegistryText	Required	1	*	Describes the relationship (Caller, Victim, suspect, etc.) of a person to the Incident. Available person types are contained in the IANA "EIDO-Person Role" registry. Note that there could be multiple relationships such as when the reporting party is also the victim.
callIdentifier	Conditional: Must be provided if this person was involved in a call	0	*	Call Identifier of the calls this person was involved in. Must be an identifier as defined in the Call Identifier section of NENA-STA-010 [4].
locationReference	Optional	0	*	Reference to a Location Data Component. Location of the person.
destinationReference	Optional	0	1	Reference to a Location Data Component. Location where the person is going. When the destination is a place to meet responders, the location type is set to "Staging".
eta	Optional	0	1	Timestamp of when the person is expected to reach their destination.

JSON Name	Use	Min	Max	Description
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the person.
ncPersonComponent	Optional	0	1	NIEM Core Person Type, a data type for a human being.
additionalDataReference	Optional	0	*	Additional Data about a person.
callBackReference	Conditional: Required if available	0	1	Reference to a Callback Data Component. Identifies how the person described in this Data Component can be called back.
personCallLegState	Optional	0	*	Call Leg State Data Component. Used to represent the person call leg state when it is different from the global call state. SHOULD be omitted if the state of the call leg is the same as the call.
itemReference	Optional	0	*	Reference to an Item Data Component. Represents an item that the person carries or has direct access to.

3.21 Organization Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is used to exchange information about organizations associated with an Incident. The information is provided by reporting parties and emergency responders. Responders can enter the information either directly through their mobile data computers or via their assigned dispatchers.

An instance of an Organization Data Component MAY be added to an EIDO and referenced by a Call Data Component and/or Incident Data Component if an organization, such as a business, is involved with the call/incident. When initially referenced by a Call Data Component, at the end of the call, while the EIDO may no longer include a Call Data Component, the organization's details will continue to be found in the Organization Data Component.

Table 3-20 Organization Data Component

JSON Name	Use	Min	Max	Description
organizationIncidentRoleRegistryText	Required	1	*	Describes the relationship (Caller, victim, suspect) of an organization to the Incident. Values MUST come from the IANA "EIDO-OrganizationIncidentRole" registry.
locationReference	Optional	0	*	Reference to a Location Data Component. Location of the organization.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the organization.
ncOrganizationComponent	Optional	0	1	NIEM Core Organization Type, a data type for a unit which conducts some sort of business or operations.
additionalDataReference	Optional	0	*	Additional Data about an organization.
callBackReference	Conditional: Required if available	0	1	Reference to a Callback Data Component. Identifies how the organization described in this Data Component can be called back.

3.22 Vehicle Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is used to exchange information about vehicles associated with an Incident including: suspect vehicles, vehicles involved in accidents, and other vehicles involved with the Incident. The information is provided by reporting parties and emergency responders. Responders can enter the information either directly through their mobile data computers or via their assigned dispatchers.

Note: Vehicle telematics information is not located in this Data Component. It is located in additional data associated with a call.

Table 3-21 Vehicle Data Component

JSON Name	Use	Min	Max	Description
vehicleRelationshipType	Required	1	*	Describes the relationship (victim's vehicle, accident vehicle, suspect vehicle, etc.) of a vehicle to the Incident. Available vehicle relationship types are contained in the IANA "EIDO-VehicleRelationshipType" registry.
vehicleRelationshipTimeStamp	Required	1	1	The date and time that the relationship of the vehicle to the Incident was established by this Data Component instance. SHALL be represented by the "date-time" datatype described in RFC 3339, Section 5.6, as specified in NENA-STA-010 [4].
additionalDataReference	Optional	0	*	Reference to an Additional Data Data Component. Additional information about a vehicle that is involved in or related to the Incident. There may be multiple data providers for one vehicle.
locationReference	Optional	0	*	Reference to a Location Data Component. Location of the vehicle.
destinationReference	Optional	0	1	Reference to a Location Data Component. Location where the vehicle is going. When the destination is a place to meet responders, the location type is set to "Staging".
eta	Optional	0	1	Timestamp of when the vehicle is expected to reach its destination.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the vehicle.
ncVehicleComponent	Optional	0	1	NIEM Core VehicleType. A data type for a conveyance designed to carry an operator, passengers, and/or cargo, over land.

JSON Name	Use	Min	Max	Description
jCommercialVehicleComponent	Optional	0	1	NIEM JXDM CommercialVehicleType. A data type for a class of vehicle that includes motor vehicles with a gross vehicle weight rating of 26,001 pounds or more.
itemReference	Optional	0	*	Reference to an Item Data Component. Represents an item that is in or on the vehicle.

3.23 Location Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component represents a location description. The type of location may be the caller's location, the Incident's location, or another type of location indicated by the Location Type field in the Data Component. In order to dispatch emergency responders, an "initial" Incident location is required. It may be the same as the caller's location, but it may also evolve as the Incident progresses. For example, emergency responders are initially dispatched to the caller's location, the caller verbally describes a different location for the Incident, and finally, the first responders arrive at the scene and relate yet another location for the Incident. The Incident's location may also be mobile such as a caller reporting an Incident from a moving vehicle or a law enforcement chase in progress.

The Location Data Component references or includes a PIDF-LO [11] [13] as documented in NENA-STA-010 [4]. When expressing a location in a civic format, the PIDF-LO by itself is unable to describe either a cross street or an intersection. In this document, the terms intersection and cross street are described as follows:

- An intersection is a location that is described as the convergence of more than one street, regardless of the orientation of the streets to each other. This includes perpendicular, T-shaped, Y-shaped, and any other orientation of streets. Intersections involve at least two streets but may include more than two streets. The location is the intersection itself.
- A cross street is an additional piece of information about a location, and describes the nearest streets that cross or terminate near the location. The cross street can have a perpendicular, T-shaped, Y-shaped, or any other orientation of the cross

street to the street the location is on. The cross street is not the location: it is near the location. There could be more than one cross street if a location is in or near the middle of a block, or the cross street has multiple street names (such as when a street changes names on either side of the street the location is on).

To represent intersections and cross streets, additional PIDF-LOs are used. The PIDF-LO SHOULD have country, A1, A2, A3 and/or A4, and MAY have A5 fields. It MUST have a street name. Implementations MUST be prepared to receive a cross street or an intersection PIDF-LO that does not contain one or more of country, A1–A5. If they are missing, they are assumed to be the same as the values in the locationByValue or referenced by the locationByReferenceUrl PIDF-LO. For an intersection, address number MAY be provided. If present, it is interpreted as a block number. For a cross street, address number MUST be provided. The PIDF-LO used in intersection and cross street adopts the field definition of CLDXF-US or CLDXF-CA but does not necessarily conform to the construction rules in those documents.

Multiple instances of intersecting streets may be present, some by value and others by reference URL. A single instance of an intersecting street MUST NOT be represented both by value and by reference. The same applies to cross street.

Cross street and intersection are mutually exclusive. If one is provided, the other MUST NOT be provided.

The elements locationByValue and locationByReferenceUrl are mutually exclusive. If one is provided, the other MUST NOT be provided. The location of a caller MUST be forwarded in the same way it was included in the call. If caller location is provided by reference, the Agency MUST provide that same URI in the EIDO. An Agency MAY report the location of an Incident (including a location it got in an EIDO) by value or by reference as determined by that Agency. The location of the caller and the location of the Incident might be the same or different, depending on the Incident.

Table 3-22 Location Data Component

JSON Name	Use	Min	Max	Description
locationTypeDescriptionRegistryText	Required	1	1	Location type (Caller, Initial, CurrentIncident, Staging, Investigation, Tower Location, Other) as defined in the IANA “EIDO-LocationType” registry.
locationByValue	Optional	0	1	The Location Data Component MUST support all PIDF-LO Data Elements though many of these elements may not be present in an EIDO. When populated, this Data Element MUST

JSON Name	Use	Min	Max	Description
				contain civic or geodetic location elements.
locationByReferenceUrl	Optional	0	1	A URL that can be dereferenced to obtain the location of the indicated location type. The resulting dereference MUST support the PIDF-LO Data Elements defined in Location By Value. This is particularly useful for indicating the location of moving devices such as callers in moving vehicles. The current location of the device can be de-referenced and inserted into the Incident record.
locationDereferencedFromReference	Conditional: MUST be populated if the content of locationByValue was obtained by dereferencing a reference URL	0	1	If locationByValue came from a dereference operation, this contains a reference to the Location Data Component containing the source locationByReferenceUrl.
locationDescriptionText	Optional	0	1	Optional text further describing the location type. Note that the Location may be the Caller's location, Incident's location, or another type of location, depending on the Location Type field.
crossStreetByValue	Optional	0	*	The nearest cross street to the Incident's location in PIDF-LO format.
crossStreetByReferenceUrl	Optional	0	*	The URL of the nearest cross street to the Incident's location.
intersectingStreetByValue	Optional	0	*	The nearest intersection to the Incident's location in PIDF-LO format.
intersectingStreetByReferenceUrl	Optional	0	*	The URL of the nearest intersection to the Incident's location.

JSON Name	Use	Min	Max	Description
cellTowerSectorId	Optional	0	1	Text field containing the ID of the nearest cell tower and the sector/face of the tower receiving the call. May be used with the "Provided By" field of the PIDF-LO to identify the carrier if carrier specific data is needed.
additionalDataReference	Optional	0	*	Reference to an Additional Data Data Component. Contains additional data associated with a location.
localIdentifier	Optional	0	*	Identifier generated by the Agency that shared the location that refers to this location in their local systems/databases.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the location.

3.24 Additional Data Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is optional and is used to exchange information about individuals, the call, or the location associated with a call received by an Agent handling the Incident. This Data Component is used to track and exchange this information as defined in RFC 7852 [10].

If additional data is provided by reference, the Agency MUST provide that same URI in the EIDO. An Agency MAY add more additional data (including data it got in an EIDO) by value or by reference as determined by that Agency. The additional data added by an Agency might include some or all of the original additional data obtained by reference.

Table 3-23 Additional Data Data Component

JSON Name	Use	Min	Max	Description
additionalDataByReferenceUrl	Conditional:	0	1	The URL is a link to an additional data block

JSON Name	Use	Min	Max	Description
	MUST be populated if additionalDataDetail is not populated. Otherwise, it MUST be omitted.			received about the parent. Allow the dereferencing of additional information associated with a call related to the Incident.
additionalDataByValue	Conditional: MUST only be populated if additionalDataByReferenceUrl is not populated. Otherwise, it MUST be omitted.	0	1	If an additional data block associated with a call is sent by value, this field contains the information.
additionalDataDereferencedFromReference	Conditional: MUST be populated if the content of additionalDataByValue was obtained by dereferencing a reference URL	0	1	If additionalDataByValue came from a dereference operation, this contains a reference to the Additional Data Data Component containing the source additionalDataByReferenceUrl.
urlPurpose	Conditional: MUST be populated if a purpose has been sent along with the URL	0	1	Purpose that accompanied the URL. For example, the purpose parameter sent with the URL in the Call-Info headers.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains notes and comments related to the Additional Data that were entered by Agents and emergency responders.
cautionReference	Optional	0	*	Reference to a Caution Data Component. Used to raise special

JSON Name	Use	Min	Max	Description
				attention to circumstances related to this Additional Data that might threaten the safety of people or property.

3.25 Emergency Resource Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: A responder can be a vehicle, a person (foot patrol), an organizational unit such as a squad or strike team, or other emergency responder configurations. A responder is described by a unique unit ID and unit type. There may be multiple Emergency Resource Data Components when multiple emergency responders are dispatched to a single Incident. When responders are assigned to an Incident by a dispatcher, then the parent Data Component is the Dispatch Data Component, and the same Agency and Agent that entered the information contained in the Dispatch Data Component entered the information contained in the Emergency Resource Data Component. However, when responders update their status or change the Incident through their Field Responder Client (FRC), then the parent Data Component is the EIDO Data Component. Agent Data Components references of Emergency Resource Data Component instances identify the individuals associated with the emergency response unit; for example, Officers Jeff Smith and John Jones are currently operating (riding in) police unit number 52.

In this Data Component, if the value “Other” is used in the resourceAttributeRegistryText field, then the resourceAttributeInternalText MUST be populated.

In this Data Component, the identifier ($id) is the unit ID. Format of the identifier is defined as unit@AgencyId (e.g., Fire1@riversideFD.riverside.ca).

This Data Component can also be used to give information about emergency resources that are being requested using EDXL-RM or other means outside the scope of this document. The unit ID of a requested resource might not be known yet. In those cases, the format of the identifier is defined as emergencyResourceTypeCommonRegistryText:requestId@AgencyId, where requestId MUST be unique for a request for the same type of resource from the same Agency (e.g., BrushFireEngineType3:1@riversideFD.riverside.ca).

Table 3-24 Emergency Resource Data Component

JSON Name	Use	Min	Max	Description
emergencyResourceTypeCommonRegistryText	Required	1	1	A standard code for an emergency type (FireTruckLadder, PatrolUnit, etc.). Restricted to values in the IANA "EIDO-EmergencyResourceType-Common" registry.
emergencyResourceTypeInternalText	Optional	0	1	A local code for an emergency resource type.
resourceAttributeRegistryText	Required	1	*	A standard code for an emergency resource attribute (skill and equipment) possessed by an emergency resource. Restricted to values in the IANA "EIDO-ResourceAttribute" registry (see Section 4.11 "EIDO-ResourceAttribute" Registry).
resourceAttributeInternalText	Conditional: MUST be populated if the value "Other" is used in the resourceAttributeRegistryText field	0	*	A local code for an emergency resource attribute.
emergencyResourceName	Optional	0	1	Common name of the emergency resource.
primaryUnitStatusRegistryText	Required	1	1	The common, globally unique status that sets the emergency resource's ability to be assigned to an emergency Incident. An emergency resource can only have one Primary Unit Status-Common at any given time. Available options for Primary Unit Status-Common are contained in the IANA "EIDO-PrimaryUnitStatus-Common" registry. Agencies should map their Unit Status-Internal to the most

JSON Name	Use	Min	Max	Description
				appropriate combination of Primary Unit Status-Common and Secondary Unit Status-Common available in the registry.
secondaryUnitStatusRegistryText	Required	1	*	Globally unique status that further qualifies the Primary Unit Status by providing more detail about the associated Primary status. Some systems may not be able to handle a Secondary status, which is acceptable, but not recommended. Available options for Secondary Unit Status-Common are contained in the IANA "EIDO- SecondaryUnitStatus-Common" registry. Agencies should map their Unit Status- Internal to the most appropriate combination of Primary Unit Status-Common and Secondary Unit Status-Common available in the registry.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains notes and comments related to the status of an emergency responder (e.g., time that status is expected to change).
unitStatusInternal	Optional	0	*	Local or internal status of a response unit. May be meaningful only to the owning Agency and possibly to other closely affiliated agencies. Some systems may not be able to handle multiple Unit Status-Internal.
agentReference	Required	1	*	Reference to an Agent Data Component. Identifies the Agents currently staffing the emergency responder unit and the Agency to which the unit belongs. If the responding unit updated the Incident directly on their FRC, then the Agent role identifies which individual entered the information as well as

JSON Name	Use	Min	Max	Description
				their role in the emergency response unit (driver, passenger, etc.).
unitLocationReference	Optional	0	1	Reference to a Location Data Component. Valid location of the unit at the time indicated by the Date/Time Stamp.
unitDestinationReference	Optional	0	1	Reference to a Location Data Component. Location where the Emergency Resource is going. When the destination is not the Incident's location, the location type is set to "Staging".
eta	Optional	0	1	Timestamp of when the unit is expected to reach its destination
personReference	Optional	0	*	Reference to a Person Data Component. Contains person information entered or updated directly by an emergency responder.
vehicleReference	Optional	0	*	Reference to a Vehicle Data Component. Contains vehicle information entered or updated directly by an emergency responder. Represents the vehicle(s) used by the Emergency Resource.
itemReference	Optional	0	*	Reference to an Item Data Component. Contains item information entered or updated directly by an emergency responder. Represents the item(s) used by the Emergency Resource (e.g., weapons, tools).
activeOnIncidentId	Optional	0	1	Incident the resource is currently active on. If the unit is not active on any Incident this Data Element will not be present. Version 1 EIDOs exclude this Data Element so it MUST be assumed the resource is active on the Incident being represented in the version 1 EIDOs.

JSON Name	Use	Min	Max	Description
stackedOnIncidentId	Optional	0	*	Incident the resource is queued on. The order of these Incident IDs is out of scope for this document.

3.26 Unit Emergency State Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is used to represent an emergency state reported for a unit. For example, an emergency button being pressed on a radio or a vehicle reporting a severe crash.

Table 3-25 Unit Emergency State Data Component

JSON Name	Use	Min	Max	Description
severity	Required	1	1	Represent the degree to which the state should be taken seriously. Estimated by the sending Agent or Agency. Possible values in order of severity are "lowest", "low", "normal", "high", and "highest." Default to "highest" if not defined.
originatingDevice	Optional	0	1	User friendly alias for the device used to request/enter Unit Emergency State. An example could be a specific radio device such as Radio1334.
emergencyStateType	Optional	0	1	Type of emergency state. Types are listed in the IANA "UnitEmergencyStateType" registry (see Section 4.8 "EIDO Unit Emergency State Type" Registry for the registry description).
reportingAgentReference	Optional	0	1	A reference to an Agent Data Component representing the Agent who triggered initiation of the Unit Emergency State from the originating device. This Agent might

JSON Name	Use	Min	Max	Description
				also be referenced by instances of the Emergency Resource Data Component, indicating the unit/resource the Agent is a member of.
terminatingAgentReference	Optional	0	1	A reference to an Agent Data Component representing the Agent who terminated the Unit Emergency State.
startTimestamp	Required	1	1	Indicates the beginning of the Unit Emergency State. SHALL be represented by the "date-time" datatype described in RFC 3339, Section 5.6, as specified in NENA-STA-010 [4].
stopTimestamp	Optional	0	1	Indicates the end of the Unit Emergency State. SHALL be represented by the "date-time" datatype described in RFC 3339, Section 5.6, as specified in NENA-STA-010 [4].
acknowledgementTimestamp	Optional	0	1	Indicates the moment an Agency acknowledges the emergency alarm. SHALL be represented by the "date-time" datatype described in RFC 3339, Section 5.6, as specified in NENA-STA-010 [4].
notesReference	Optional	0	*	Reference to a Notes Data Component.

3.27 Alarms and Sensors Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is used to support the exchange of the legacy interface described in the *Alarm Monitoring Company to Emergency Communications Center (ECC) Computer–Aided Dispatch (CAD) Automated Secure Alarm*

Protocol (ASAP), APCO/TMA 2.101 [12], and other alarm data. There may not be an i3 call associated with this alarm. The Incident Record Handling FE will have a direct interface that supports APCO/TMA 2.101 [12], and the FE will use it to automatically extract the relevant information and create an Incident. This Data Component provides a link to the original information received from the alarm company. Any other form of alarm or sensor data will be contained in the Additional Data Associated with a Call [10]. Future versions of this document may consider updated versions of the APCO/TMA interface [12].

Table 3-26 Alarms and Sensors Data Component

JSON Name	Use	Min	Max	Description
csaaAlarmInformation	Conditional: MUST be populated if alarmUrl is not populated. Otherwise, it MUST be omitted.	0	1	Original automated alarm data that triggered the creation of the Incident. Read in NIEM schema Alarm and Sensor data. For Alarms, this would be the APCO/TMA 2.101.X [12] standard.
alarmUrl	Conditional: MUST be populated if CSAAAlarmInformation is not populated. Otherwise, it MUST be omitted.	0	1	Link to the automated alarm data that initiated the Incident. There may be more than one transmission for a single Incident. Enable the receiving Agency to dereference and obtain the original alarm information that triggered the Incident.
agentReference	Optional	0	1	Reference to an Agent Data Component. Identifies the Agent and Agency that processed the Alarm/Sensor information described in this Data Component. Defaults to the Agent Information included in the EIDO Data Component, if this Data Component is not present.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the Incident.

3.28 Caution Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is optional and is used to raise special attention to circumstances related to a person, location, or vehicle that might threaten the safety of people or property.

Table 3-27 Caution Data Component

JSON Name	Use	Min	Max	Description
cautionTypeInternalText	Required	1	1	Textual description indicating the type of caution information. This is the internal text used by the local agencies involved in the Incident.
cautionTypeCommonRegistryText	Required	1	1	Type of caution as described in the IANA "EIDO-CautionType" registry (see Section 4.4 "EIDO Caution Type" Registry for the registry description).
cautionTypeCommonSubRegistryText	Optional	0	*	Registry values further describing the caution. More than one is allowed when they are related. Standard documents are allowed to ask IANA to create new subregistries.
cautionSeverity	Required	1	1	Represents the degree to which the caution should be taken seriously. Estimated by the sending Agent or Agency. Possible values in order of severity are "lowest", "low", "normal", "high", and "highest."
cautionSource	Optional	0	1	Person, Agent or Agency that provided the caution information. If the information is provided by an Agent with a known Agent Id, then the Agent Id must be the value. If the information is provided by an Agency with a known Agency Id, then the Agency Id must be the

JSON Name	Use	Min	Max	Description
				value. If the information is provided by a person referenced in the EIDO, then the Person Data Component Identifier ($id) must be the value. Otherwise, the value is free form text describing the source.
cautionSourceType	Optional	0	1	Describe the type of value in the cautionSource Data Element. Possible values are "AgentId", "AgencyId", "PersonId" or "Text".
startDate	Optional	0	1	Date and time caution was first registered.

3.29 Item Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Emergency Incident Data Object

Data Component Description: This Data Component is used to represent items that are of interest in the context of the Incident. An Item can be something like a firearm (including a specific type) or other weapon, medical equipment, jewelry, a credit card, sports equipment, musical instruments, etc. The list of valid Item types is very broad and is adopted from the NIEM ItemType.

Implementations SHOULD NOT use Item Data Component for items that have a dedicated Data Component defined in this document where NIEM provides a valid entry for that item. For example, NIEM's ItemType incorporates VehicleType, so it is technically valid to use Item to describe a vehicle. However, because EIDO has a dedicated Vehicle Data Component, implementations should use that Data Component to describe a vehicle, and should not use Item.

Table 3-28 Item Data Component

JSON Name	Use	Min	Max	Description
ncItemType	Optional	0	1	NIEM Core ItemType. A data type for an article or thing. Many NIEM types extend the ItemType and can be represented in

JSON Name	Use	Min	Max	Description
				this field e.g., j:FirearmType,j:ExplosiveType, j:ContrabandType, j:JewelryType
locationReference	Optional	0	*	Reference to a Location Data Component. Location of the item.
notesReference	Optional	0	*	Reference to a Notes Data Component. Contains optional alphanumeric text further describing the item.

3.30 Alternate Values Data Component

Data Component Use: Optional

Minimum: 0

Maximum: *

Child Of: Common Prologue

Data Component Description: Subscribers may need to see conflicting values not chosen by an IDX coalescing algorithm. For this purpose, a Data Component for an EIDO is defined which contains the differences. For each Data Component where a constituent sends a candidate value that the IDX does not use in the Data Component, the IDX includes the Alternate Values Data Component. That Data Component has conflicting values from the constituents.

This Alternate Values Data Component is an optional Data Component that occurs in any Data Components where there are multiple possible values the IDX considers. The alternative values reported to the IDX for inclusion in the EIDO SHALL be included in the Alternative Data Component.

Table 3-29 Alternate Values Data Component

JSON Name	Use	Min	Max	Description
element	Required	1	1	Name of Data Element in the referenced component.
alternative	Required	1	*	Alternative value for the element.

3.30.1 Alternative Data Component

Table 3-30 Alternative Data Component

JSON Name	Use	Min	Max	Description
constituent	Required	1	1	Agency/Service/FE Id of constituent.
reportedByAgentReference	Optional	0	1	Reference to an Agent Data Component. Agent that provided the alternate value.
timestamp	Required	1	1	Timestamp of the Data Component.
otherValue	Optional	0	1	JSON object value for this element sent by constituent. If the coalescing algorithm result turns out to not update a value because a source does not include it, the EIDO MAY contain an alternate value Data Component that does not include an "OtherValue" field.

4 IANA Actions

This section describes the registries defined for the Data Components contained in Section 3 Data Associated with an Emergency Incident.

Each defined registry is linked to one or more Data Elements specified in the Data Components contained in Section 3 Data Associated with an Emergency Incident. Each registry is named after the Data Element that uses it and includes references to the Data Components that contain the registry's Data Element.

4.1 "EIDO-CommonDispositionCode" Registry

IANA is requested to add the following values to the "EIDO-CommonDispositionCode" registry in the Emergency namespace.

Value	Literal Description	Reference
101	OutOfArea—Call was closed as it occurred outside of the Agency's jurisdictional area.	This document
102	Unfounded—Call was closed as there was no evidence that a crime was committed.	This document

4.2 "EIDO Shared Communication Resource Type" Registry

IANA is requested to add a new registry in the Emergency namespace.

The "sharedCommunicationResourceType" Data Element is described in Section 3.12 Shared Communication Resource Data Component. A Shared Communication Resource is an asset that may be assigned to an Incident that assists the responders working an Incident to communicate and collaborate with each other. As an example, the radioChannel or Talk Group on a radio system is a Shared Communication Resource. Shared Communication Resources have a type. The type is a selection from this registry.

4.2.1 Name

EIDO Shared Communication Resource Type

4.2.2 Information Needed to Create a New Value

Name, description of the resource type code, and a stable URL to a schema defining a related data structure.

4.2.3 Registration policy

Specification Required [RFC 8126], which requires expert review. The expert should confirm that the entry represents a unique Shared Communication Resource type. The expert should validate that the Schema URL points to a schema that correctly represents the named resource.

4.2.4 Content

This registry contains:

- "Name" (an ASCII Token with no whitespace)
- The UTF-8 "Description" (a short string)
- "Schema URL"
- A "Reference" (URI) to a document that requests the entry

4.2.5 Initial Values

Name	Description	Schema URL	Reference
PTT-P25	Land Mobile Radio (LMR). A radio talk path, typically a P25 or analog talkgroup on a conventional or trunked radio system. However, it may be operating in direct (talk-around) mode or through a conventional radio repeater.	https://github.com/NENA911/SharedCommunicationResources/blob/main/PTT-P25.yaml	This document

Name	Description	Schema URL	Reference
	The only required field is channelAlias. All other fields are optional.		
HTTP-URL	Generic shared communication resource that can be accessed via a URL using HTTP enabled application. Must only be used if there is not a more appropriate resource type that can be used.	https://github.com/NENA911/SharedCommunicationResources/blob/main/URL.yaml	This document
SIP-TEL-URI	SIP or TEL URI used to join a conference bridge.	https://github.com/NENA911/EIDO-JSON/tree/main/Schema/?	This document
Other	Other Incident Resource.	https://github.com/NENA911/SharedCommunicationResources/blob/main/Other.yaml	This document

4.3 "EIDO Eido Extension Id" Registry

IANA is requested to add a new registry in the emergency namespace.

EIDO-EidoExtensionId registry values are meant to identify extensions to an EIDO to handle capabilities not common to all implementations. It is useful to list all such extensions and provide a way to inform others of what they are. This registry provides a way to inform others about an extension to the EIDO.

The "eidoExtensionId" Data Element is described in Section 3.5 Emergency Incident Data Object Data Component. An example Extension ID could be "WidgetSensor" with a description of "Device used in conjunction with the widget" and with a URL of "https://github.com/ThirdParty/EIDO- WidgetSensorExtension".

4.3.1 Name

EIDO Eido Extension Id

4.3.2 Information Needed to Create a New Value

ID, description of the Extension, and a stable URL to a schema defining a related data structure.

4.3.3 Registration policy

Specification Required [RFC 8126], which requires expert review. The expert should confirm that the entry represents a unique Extension. The expert should validate that the Schema URL points to a schema that correctly represents the Extension.

4.3.4 Content

This registry contains:

- "ID" (a UTF-8 Token with no whitespace)
- The UTF-8 "Description" (a short string)
- "Schema URL"
- A "Reference" (URI) to a document that requests the entry

4.3.5 Initial Values

Since this is for extending the EIDO, there are no initial values in this document.

4.4 "EIDO Caution Type" Registry

IANA is requested to add a new registry in the Emergency namespace.

Subregistries of this registry are anticipated, but this document does not create any. The "cautionTypeCommonRegistryText" Data Element is described in Section 3.28 Caution Data Component.

4.4.1 Name

EIDO Caution Type

4.4.2 Information Needed to Create a New Value

ID of the caution type, description of the Caution Type, and a "Reference" (URI) to a document that requests the entry.

4.4.3 Registration Policy

Expert Review. No standards document required. The expert should confirm that the entry represents a new Caution Type that is unique and valid.

4.4.4 Content

This registry contains:

- "ID" (an ASCII Token with no whitespace)
- The UTF-8 "Description" (a short string)
- A "Reference" (URI) to a document that requests the entry

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ID	Description	Reference
SpecialAttention	This value can be used when special attention is requested about a piece of information.	This document
Weapon	Weapon is present. Weapons include firearms, knife, or other.	This document
MedicalCondition	Condition of a person that may require customized intervention. cautionTypeCommonSubRegistryText value comes from the IANA "EIDO-MedicalCondition" registry with descriptions. These codes are not intended to be used to replace Emergency Medical Dispatch codes or protocols, standardized Incident types, or other existing medical condition type lists.	This document
HazardousCondition	Hazardous material or condition is present. This includes a chemical spill, electrocution risk, unsafe structure, etc.	This document
DangerousIndividual	Caution is related to a dangerous/violent person.	This document
Transportation	Caution is related to a transportation hazard. Examples include traffic congestion, a blocked road, redirected traffic, flooding, etc.	This document
ThreatToFacility	Threat against the location/building has been reported.	This document
FrequentCaller	Frequent emergency calls are associated with the specific device, caller, or location.	This document
Domestic	Known location where domestic violence occurred.	This document
HostileToResponder	Known hostility toward responders.	This document
Trespassing	Knowingly entering a place or property without permission.	This document
RestrainingOrder	Violation of a court restraining order.	This document
Vacancy	Location should be vacant.	This document

ID	Description	Reference
SecurityCheck	Approved Agency personnel are checking the area.	This document
Encampment	Location is known to host provisionals shelter.	This document

4.5 "EIDO Medical Condition" Registry

IANA is requested to add a new registry as a subregistry of the EIDO-CautionType registry in the Emergency namespace.

This registry contains enumerated values that each represent a broad, general medical condition of a person related to an emergency Incident, where the medical condition is ancillary to the emergency and could be provided to a responder or responding Agency for situational awareness. These codes are not intended to be used to replace Emergency Medical Dispatch codes or protocols, standardized Incident types, or other existing medical condition type lists.

4.5.1 Name

EIDO Medical Condition

4.5.2 Information Needed to Create a New Value

A "Condition" token representing the medical condition and a representative description of the medical value.

4.5.3 Registration Policy

Expert Review. No document required. The expert SHOULD confirm that the entry represents a broad type of medical condition not already represented in the registry and can be used by a responder or responding Agency for situational awareness.

4.5.4 Content

This registry contains:

- "Condition" (a short ASCII string)
- "Description" (explanatory UTF-8 text describing the meaning of the "Condition")
- "Reference" (URI that defines the value, if defined in a document)

4.5.5 Initial Values

Condition	Description	Reference
CardiacHx	Cardiac history	This document
DiabeticHx	Diabetic history	This document

Condition	Description	Reference
TbiHx	Traumatic brain injury history	This document
Blind	Blind	This document
DeafHh	Deaf/hard of hearing	This document
MhHx	Mental health history	This document
BleedingDis	Bleeding disorders	This document
NeuroDis	Neurological disorders/disease (autism, ADHD, Dementia, etc.)	This document
BreathingDis	Breathing disorders/respiratory history	This document
SeizureHx	Seizure history	This document
Mobility	Physical handicap/mobility issues (bed/wheelchair bound, paraplegic, etc.)	This document
Allergic	Allergic condition	This document
StrokeHx	Stroke history	This document

4.6 "EIDO External Information Source" Registry

IANA is requested to add a new registry in the Emergency namespace.

The external information source registry is used in the External Information Source Data Component. It represents a source of external data not defined by an additional data block that can be queried by an Agency. It is expected that multiple databases, e.g., Departments of Motor Vehicles (DMV), will be represented by their own entry in this registry.

4.6.1 Name

EIDO External Information Source

4.6.2 Information Needed to Create a New Value

Token identifying the external information source and the description of the external information source. The description should include enough information to retrieve the information from the external information source.

4.6.3 Registration policy

Expert Review. No standards document required. The expert should confirm that the entry represents an external information source not already present in the registry.

4.6.4 Content

This registry contains:

- "ID" (a UTF-8 Token with no whitespace)

- The UTF-8 "Description" (a concise string)

4.6.5 Initial Values

ID	Description
IS-ADR	Identity Searchable Additional Data Repository as defined in NENA-STA-010 [4]. The externalSourceAccessString MUST contain the full URI including the CallerURI to use with HTTP(S) GET to access the Additional Data Block(s).
IS-LIS	Identity Searchable Location Information Server as defined in NENA-STA-010 [4]. The externalSourceAccessString MUST contain the full URI, including the CallerURI to use with HTTP(S) to access the PIDF-LO.
CPIC	Canadian Police Information Centre (CPIC) is a Canadian federal database that provides information about crimes and criminals.
NCIC	National Crime Information Center (NCIC) system is a database for ready access by a criminal justice Agency making an inquiry and for prompt disclosure of information in the system from other criminal justice agencies about crimes and criminals.

4.7 "EIDO Unit Emergency State Source" Registry

IANA is requested to add a new registry in the Emergency namespace.

Values represent trigger mechanisms for a unit emergency state.

4.7.1 Name

EIDO Unit Emergency State Source

4.7.2 Information Needed to Create a New Value

Token identifying the emergency state source and the description of the emergency state source.

4.7.3 Registration policy

Expert Review. No standards document required. The expert should confirm that the entry represents an emergency state source not already present in the registry.

4.7.4 Content

This registry contains:

- "ID" (a UTF-8 Token with no whitespace)
- The UTF-8 "Description" (a concise string)

4.7.5 Initial Values

ID	Description
VehicleMobileDevice	Device usually docked in and assigned to a vehicle (e.g., ambulance, police vehicle).
Dashcam	Vehicle camera that triggers an emergency state.
VehicleSensor	Sensor in a vehicle that triggers an emergency state.
HandHeldMobileDevice	Device usually carried by an Agent (e.g., cellphone or a tablet).
Bodycam	Wearable camera typically used by first responder.
HolsterSensor	Sensor that detects that a firearm has been unholstered.
HandHeldRadioButton	A dedicated button on a handheld radio that triggers an emergency state.
VehicleRadioButton	A dedicated button on a unit vehicle radio that triggers an emergency state.
VehicleRadioVocal	An Agent used a vehicle radio to vocally trigger an emergency state.
HandHeldRadioVocal	An Agent used a handheld radio to vocally trigger an emergency state.

4.8 “EIDO Unit Emergency State Type” Registry

IANA is requested to add a new registry in the Emergency namespace.

Values represent the emergency type for a unit emergency state.

4.8.1 Name

EIDO Unit Emergency State Type

4.8.2 Information Needed to Create a New Value

Token identifying the emergency state type and the description of the emergency state type.

4.8.3 Registration policy

Expert Review. No standards document required. The expert should confirm that the entry represents an emergency state type that is not already present in the registry.

4.8.4 Content

This registry contains:

- "ID" (a UTF-8 Token with no whitespace)
- The UTF-8 "Description" (a concise string)

4.8.5 Initial Values

ID	Description
responderDown	Used to indicate that a responder's life is in danger.
weaponDrawn	Used to indicate a responder has a weapon drawn.

4.9 "LogEvent CallTypes" Registry

IANA is requested to add the following Values to the "LogEvent CallTypes" registry in the Emergency namespace.

Value	Literal Description	Classification	Reference
nonInteractive	The call is a non-interactive call as described in RFC 8876 [8].	Secondary	This document
callback	A Callback is the action of reconnecting to an emergency caller after the Emergency Call has ended.	Primary	This document

4.10 "EIDO-SecondaryUnitStatus-Common" Registry

IANA is requested to add the following values to the "EIDO-SecondaryUnitStatus-Common" registry in the Emergency namespace.

Value	Literal Description	Reference
Requested	A pending request for resource was made for this Emergency unit.	This document
ResponderNeedAssistance	Responder advises they need assistance.	This document
ResponderInTrouble	Responder advises they are in immediate danger.	This document

4.11 "EIDO-ResourceAttribute" Registry

IANA is requested to add the following values to the "EIDO-ResourceAttribute" registry in the Emergency namespace.

Name	Description	Reference
Other	The Emergency Resource has attributes not listed elsewhere in this registry. If this value is used in the resourceAttributeRegistryText field, then the resourceAttributeInternalText MUST be populated.	This document

4.12 "EIDO Organization Incident Role" Registry

IANA is requested to add a new registry in the Emergency namespace.

Describes a relationship (Caller, victim, suspect) of an organization in the context of an Emergency Incident.

4.12.1 Name

EIDO Organization Incident Role

4.12.2 Information Needed to Create a New Value

Token identifying the organization role and the description of the organization role in the context of an emergency Incident. The description should be in clear text and describe the role.

4.12.3 Registration policy

Expert Review. No standards document required. The expert should confirm that the entry is not already present in the registry and represents a potential role of an organization in the context of an emergency Incident.

4.12.4 Content

This registry contains:

- "ID" (a UTF-8 Token with no whitespace)
- The UTF-8 "Description" (a concise string)

4.12.5 Initial Values

Value	Literal Description
InvolvedOrganization	Organization described in the Organization Data Component that is involved in the Incident. Used when no other relationship is known.
CallOriginator	An emergency call that originated from the Organization described in the Organization Data Component.

Value	Literal Description
ReportingParty	Organization described in the Organization Data Component that is involved in the Incident as a reporting party.
Victim	Organization described in the Organization Data Component that is involved in the Incident as a victim.
Assisting	Organization described in the Organization Data Component that is assisting with the Incident in some way (e.g., provide data about the Incident, picture, video).
Subject	The Organization that is the subject of the reported Incident.

5 Impacts and Considerations

The FEs exchanging the data defined in the EIDO may be physically or virtually connected to each other. The FEs may belong to one or more disparate manufacturers' systems located within the same public safety Agency or within different agencies.

5.1 Operations Impacts Summary

A standardized format for electronically exchanging emergency Incident information will provide stakeholders with many operational benefits. These benefits are similar to those realized when agencies implement a local or regional "CAD to CAD" exchange. Use of the EIDO at a national, regional, and local level and within communication centers themselves is critical to the implementation of NG9-1-1 (i3) compliant systems [4].

5.2 Technical Impacts Summary

As with the implementation of any technical standard, the EIDO will have a significant impact to emergency services. Initially, all FEs involved in the exchange of emergency Incident information (e.g., call handling, logging, dispatch) will need to be modified to comply with EIDO transactions. Minimal impacts are expected on the ESInet or other E9-1-1 IP-based networks; however, they will need to be compatible with the EIDO structure in order to carry EIDOs from one FE to another.

5.3 Security Impacts Summary

EIDOs carry confidential information, and are transmitted over secure transport such as TLS. EIDOs accessible to authenticated FEs and other systems can have their contents filtered to contain only data authorized to be transmitted to those systems by the data owner's policy. Furthermore, FEs and systems that pass EIDOs or their contents along to other FEs and systems can filter those EIDO instances based on the authentication of the receiving FEs and other systems to contain only data authorized to be transmitted to those FEs and systems by the data owner's policy.

5.4 Recommendation for Additional Development Work

Future versions may expand the EIDO to include new Data Components and Data Elements, provide additional allowable registry values, or be modified to support additional emergency Incident related exchanges such as the transfer of a patient's medical diagnostics, administered procedures, and medical status between transporting ambulances and receiving hospitals.

Table 5-1 Additional Development Work

Section(s)	Reference to future work
2, Appendix A	Adding NIEM Injury treatment type.
Appendix A	Conformance with future NIEM JSON version.
TBD	Add a mapping between EIDO and NIEM em:EmergencyIncidentType.
3.1	Make TransferredTo Agency role transient. Should an Agency keep that role once the transfer is completed or when the call is then transferred to another Agency? The working group recommends referring this to the NG-PSAP working group since this business rule would be out of scope for the data structure itself.
2.7,2.8	Add reason code for Merge Data Component and Link Data Component.
2.9	Add information about how an Incident is originally reported.
2.x	Add URL for media related to an Incident (e.g., video feed).
2.7	Review allowed values in incidentMergeDirectionCode. Possibly use REPLACED BY rather than REPLACED, and REPLACES rather than REPLACING. Possibly use RETIRED and SURVIVING as the values.
2.7	Add new value to incidentMergeDirectionCode to represent unmerge operation.
3.2	Add a PTT-TETRA value to the EIDO-SharedCommunicationResourceType registry and create a corresponding schema.
3.2	Add a MCPTT value to the EIDO-SharedCommunicationResourceType registry and create a corresponding schema.
	APCO 2.103 [7] should be updated to match with the IANA “EIDO-IncidentType-Common” registry.
2	Consider the potential need for a flag for Data Components to apply XACML policies (e.g., Law enforcement, medical information).
2	Consider being able to represent the source (Agent, Agency or device) of Data Elements.
2	Consider updated versions of the APCO CSAA ANS interface in the Alarms and Sensors Data Component.

Section(s)	Reference to future work
3	Add a way to specify who documented each data element.
3	Add a Correlator element or component to specify a connection between reported information, for example these 2 cars are likely the same.
3	Normalize the usage of the "timestamp" term in Data Elements so it is considered a single word by not capitalizing the "s".
-	Add additional validation to the schema in version 2. See pull request https://github.com/NENA911/EIDO-JSON/pull/34

5.5 Anticipated Timeline

EIDO is an integral piece of NG9-1-1 and must be considered during implementation planning. Early implementations are anticipated considering the migration to NG9-1-1 in Canada.

5.6 Cost Factors

Significant consideration was given to the cost impact of proposing yet another "change" to 9-1-1 stakeholders. Standardizing the format of Incident information exchange, however, is critical for NG9-1-1's true potential to be realized [4]. Absent this standard, Incident information data exchanges will continue to rely on limited, proprietary implementations that are costly and offer limited capability to share Incident information both within and between communication centers.

It should be noted that implementation of the EIDO IEPD ANS, as well as most NG9-1-1 systems, may require updates to existing public safety systems resident at communication centers.

5.7 Cost Recovery Considerations

Normal business practices shall be assumed to be the cost recovery mechanism.

5.8 Additional Impacts (non-cost related)

The information or requirements contained in this document are not expected to have additional impacts, based on the analysis of the authoring group.

6 Abbreviations, Terms, and Definitions

See the NENA Knowledge Base (NENAkb) Glossary [1] for a list of terms and abbreviations used in NENA documents. Abbreviations and terms used in this document are listed below with their definitions as they apply to this version of the document. Please be aware that terms and definitions in the NENAkb Glossary may be different than what is shown in this document.

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
Callback Call	A Callback is the action of reconnecting to an Emergency Caller after the Emergency Call has ended where the attempt to reconnect occurs within a configurable amount of time of its reception. Callback calls have special signaling.	Add
Incident	A real-world occurrence, or event, for which one or more calls may be received.	In Glossary
Agency	An Agency is an entity with a valid public safety purpose under a single discrete recognized administration. In 9-1-1 and public safety operations, an Agency is a governmental entity, or a non-governmental entity under the direction of a governmental entity, responsible for all or some part of 9-1-1 system provisioning, call processing, and field response.	Do not update
Data Component	A structure composed of multiple Data Elements.	Don't Add
Data Element	A unit of information composed of a name and value. The value can be a number, a boolean, a character string, or a Data Component.	Don't Add
EIDO (Emergency Incident Data Object)	A JSON-based (JavaScript Object Notation) object that is used to share emergency Incident information between and among authorized entities and systems. NENA has adopted the JSON-based EIDO for sharing Incident information among authorized NG9-1-1 entities and systems.	In Glossary
ESInet (Emergency Services IP Network)	An ESInet is a managed IP network that is used for emergency services communications, and which can be shared by all public safety agencies. It provides the IP transport infrastructure upon which independent application platforms and core services can be deployed, including, but not restricted to, those necessary for providing NG9-1-1	In Glossary

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
	services. ESInets may be constructed from a mix of dedicated and shared facilities. ESInets may be interconnected at local, regional, state, federal, national, and international levels to form an IP-based internetwork (network of networks). The term ESInet designates the network, not the services that ride on the network. See NGCS (Next Generation Core Services)	
FE (Functional Element)	An abstract building block that consists of a set of interfaces and operations on those interfaces to accomplish a task. Mapping between functional elements and physical implementations may be one-to-one, one-to-many, or many-to-one. **Also known as:** ***Functional Entity***	In Glossary
Follow-Up Call	A call to a previous 9-1-1 caller after the Callback Period Timer has elapsed and while the Incident is active.	Add
FRC (Field Responder Client)	A device or software installed on a device such as a computer, tablet, or smartphone used by field responders to access public safety systems, such as to receive dispatch instructions or perform database lookups. FRCs are often colloquially referred to as "MDT (Mobile Data Terminal)."	In Glossary
IMR (Interactive Media Response)	An automated service used to play announcements, record responses, and interact with callers using any or all of audio, video, and text.	In Glossary
IS-ADR (Identity Searchable Additional Data Repository)	An Additional Data Repository that provides a service that can search for Additional Data based on a sip/sips or tel URI: (e.g., Additional Data about the caller).	In Glossary

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
MDT (Mobile Data Terminal)	A computerized device used in a vehicle to exchange information between an end user and a communications center. MDTs are a specific type of FRC (Field Responder Client).	In Glossary
P-Asserted-Identity	A header field in a SIP message containing a URI that the originating network asserts is the correct identity of the caller. **Also known as:** ***PAI***	In Glossary
Report Number	A number associated with an emergency Incident that once generated indicates that a follow up report or investigation will be associated with the Incident. May also be known as other variations depending on the types of service agencies involved and local customs. Typically, the report number is sequential within a year and also identifies the Agency for which it is issued. Within an Agency, the report number is globally unique. More than one report number may be associated with a single Incident. **Also known as:** ***Case number*** ***Fire Incident number*** ***Department report number***	Add
SIP (Session Initiation Protocol)	A protocol specified by the IETF (RFC 3261) that defines a method for establishing multimedia sessions over the internet. Used as the call signaling protocol in VoIP, NENA I2, and NENA i3.	In Glossary
TCP (Transmission Control Protocol)	The reliable and connection-oriented octet streaming transport protocol of the Internet protocol suite. It is specified by the IETF in RFC 9293.	Add

Term or Abbreviation (Expansion)	Definition	Recommendation for NENAkb Glossary
TLS (Transport Layer Security)	An IETF standard protocol that provides authentication, integrity protection, and confidentiality using mutually negotiated cryptographic capabilities and digital certificates on a connection such as TCP. It is specified by the IETF in RFC 8446.	Update
URL (Uniform Resource Locator)	A URL is a type of URI, specifically used for describing and navigating to a resource (e.g., https://www.nena.org)	In Glossary

7 References

[1] National Emergency Number Association. "NENA Knowledge Base Glossary." Updated May 6, 2025. https://kb.nena.org/wiki/Category:Glossary.

[2] Internet Engineering Task Force. *Key words for use in RFCs to Indicate Requirement Levels. S. Bradner*. RFC 2119, March 1997.

[3] OpenAPI Initiative. *OpenAPI Specification 3.0.2 Released*. https://spec.openapis.org/oas/v3.0.2.html, October 08, 2018.

[4] National Emergency Number Association. *NENA i3 Standard for Next Generation 9-1-1*. NENA-STA-010.3-2021. Alexandria, VA: NENA, approved July 12, 2021.

[5] NIEM Program Management Office. *Introduction to the National Information Exchange Model (NIEM)*, version 0.3. Issued February 12, 2007. https://reference.niem.gov/niem/guidance/introduction/0.3/niem-introduction-0.3.pdf.

[6] Internet Engineering Task Force. *A Uniform Resource Name (URN) for Emergency and Other Well-Known Services*. H. Schulzrinne. RFC 5031, January 2008.

[7] Association of Public-Safety Communications Officials International. *Public Safety Communications Common Incident Types for Data Exchange*. APCO 2.103.2-2019.Daytona Beach, FL: APCO, approved October 18, 2019.

[8] Internet Engineering Task Force. *Non-interactive Emergency Calls*. B. Rosen, H. Schulzrinne, H. Tschofenig, and R. Gellens. RFC 8876, September 2020.

[9] National Emergency Number Association. *NENA NG9-1-1 PSAP and ECC Specifications for the NENA i3 Solution*. NENA-STA-023.1-202Y. Alexandria, VA: NENA (forthcoming).

[10] Internet Engineering Task Force. *Additional Data Related to an Emergency Call.* R. Gellens, B. Rosen, H. Tschofenig, R. Marshall, and J. Winterbottom. RFC 7852, July 2016.

[11] Internet Engineering Task Force. *GEOPRIV Presence Information Data Format Location Object (PIDF-LO) Usage Clarification, Considerations, and Recommendations.* J. Winterbottom, M. Thomson, and H. Tschofenig. RFC 5491, March 2009.

[12] Association of Public-Safety Communications Officials International. *Alarm Monitoring Company to Emergency Communications Center (ECC) Computer – Aided Dispatch (CAD) Automated Secure Alarm Protocol (ASAP)*. APCO/TMA 2.101.3-2021. Daytona Beach, FL: APCO, approved May 21, 2021.

[13] Internet Engineering Task Force. *A Presence-based GEOPRIV Location Object Format*. J. Peterson. RFC 4119, December 2005.

[14] National Emergency Number Association *NENA Rules for Use of Version Control Repositories*. NENA-ADM-012, November 2022 (update pending).

[15] World Wide Web Consortium (W3C). *W3C XML Schema Definition Language (XSD) 1.1 Part 1: Structures*. S. Gao, C. M. Sperberg-McQueen, and H. Thompson. https://www.w3.org/TR/xmlschema11-1/, April 5, 2012.

[16]

8 Appendices

8.1 Appendix A—Required NIEM Data Elements that Need to be Supported for EIDO Compliance

The Person Data Component and the Vehicle Data Component both contain their respective NIEM Data Components. The NIEM PersonType and VehicleType also contain other NIEM Data Elements and Data Components.

In order for implementations to be interoperable, a subset of these NIEM Data Elements and Data Components must be supported by every EIDO implementation. In this context, support means that the Data Element must be preserved when transmitting, receiving, storing, or retrieving an EIDO.

To be compliant with this standard, any EIDO implementation must support the following NIEM Data Elements.

8.1.1 Person Type

Table 8-1 Person Type

JSON Name	NIEM Data Element	Type	Description	Comment
ageMeasure	nc:PersonAgeMeasure	nc:TimeMeasureType	A measurement of the age of a person.	
birthDate	nc:PersonBirthDate	nc:DateType	A date a person was born.	
birthLocation	nc:PersonBirthLocation	nc:LocationType	A location where a person was born.	
bloodTypeCode	j:PersonBloodTypeCode	ncic:BLTCodeType	A blood group and RH factor of a person.	
buildCode	j:PersonBuildCode	ndex:PersonBuildCodeType	A person's physique or body shape.	
citizenshipIso3166Alpha2Code	nc:PersonCitizenshipISO3166Alpha2Code	iso_3166:CountryAlpha2CodeType	A country that assigns rights, duties, and privileges to a person because of the birth or	

JSON Name	NIEM Data Element	Type	Description	Comment
			naturalization of the person in that country.	
clothing	j:PersonClothing	j:ClothingType	An article of clothing, dress, or attire for a person.	
complexion	nc:PersonComplexionText	nc:TextType	An appearance or condition of the skin of a person.	e.g., clear, freckled, wrinkled
description	nc:PersonDescriptionText	nc:TextType	A description of a person.	
digitalImage	nc:PersonDigitalImage	nc:ImageType	A photograph or image of a person in a digital format.	
disguise	nc:PersonDisguiseDescriptionText	nc:TextType	A description of something a person wears to conceal or mislead others as to the true appearance or identity of that person.	e.g., wig, mask, glasses, uniform
ethnicityCode	j:PersonEthnicityCode	ucr:EthnicityCodeType	A cultural lineage of a person.	
eyeColorCode	j:PersonEyeColorCode	ncic:EYECodeType	A color of the eyes of a person.	
eyewearCode	j:PersonEyewearCode	ndex:PersonEyewearCodeType	A description of glasses or other eyewear a person wears.	
facialHairCode	j:PersonFacialHairCode	ndex:PersonFacialHairCodeType	A kind of facial hair of a person.	
generalAppearance	nc:PersonGeneralAppearanceDescriptionText	nc:TextType	A description of the way a person looks and is presented overall.	
hairAppearance	nc:PersonHairAppearanceText	nc:TextType	An overall appearance of the hair of a person.	

JSON Name	NIEM Data Element	Type	Description	Comment
hairCategory	nc:PersonHairCategoryText	nc:TextType	A kind of hair of a person, such as wavy or straight.	
hairColorCode	j:PersonHairColorCode	ncic:HAIRCodeType	A color of the hair of a person.	
hairLengthCode	j:PersonHairLengthCode	ndex:PersonHairLengthCodeType	A length of hair of a person.	
hairStyleCode	j:PersonHairStyleCode	ndex:PersonHairStyleCodeType	A style or cut of hair worn by a person.	
handedness	nc:PersonHandednessText	nc:TextType	A hand with which a person is more adept using.	e.g., left, right, ambidextrous
height	nc:PersonHeightMeasure	nc:LengthMeasureType	A measurement of the height of a person.	
injury	nc:PersonInjury	nc:InjuryType	A form of physical harm or damage sustained by a person.	
jewelry	nc:PersonJewelryDescriptionText	nc:TextType	A description of adornments a person wears.	
isLanguageEnglish	nc:PersonLanguageEnglishIndicator	niem-xs:boolean	True if a person understands and speaks English; false otherwise.	
license	nc:PersonLicenseIdentification	nc:IdentificationType	An identification that references a license certification or registration of a person for some purpose.	This may be granted to certify a professional occupation or skill.
isLiving	nc:PersonLivingIndicator	niem-xs:boolean	True if a person is alive; false if a person is dead.	This indicator may be useful if a person is known to be

JSON Name	NIEM Data Element	Type	Description	Comment
				dead, but the death date is unknown.
unionStatusText	nc:PersonUnionStatusText	nc:TextType	A legal status of a union between two people.	
mentalState	nc:PersonMentalStateText	nc:TextType	A mental condition of a person.	
name	nc:PersonName	nc:PersonNameType	A combination of names and/or titles by which a person is known.	
nationalIdentification	nc:PersonNationalIdentification	nc:IdentificationType	An identification that references a person within a country but is not based on fingerprint.	
nationalityCode	nc:PersonNationalityISO3166Alpha2Code	iso_3166:CountryAlpha2CodeType	A country of a person's citizenship or a country in which a person is deemed a national.	
physicalFeature	nc:PersonPhysicalFeature	nc:PhysicalFeatureType	A prominent or easily identifiable aspect of a person.	
primaryLanguage	nc:PersonPrimaryLanguage	nc:PersonLanguageType	A capacity of a person for a language with which that person has the strongest familiarity.	Native language. This can also be used for bilingual or multilingual people.
raceCode	j:PersonRaceCode	ncic:RACECodeType	A classification of a person based on factors such as geographical locations and genetics.	
sexCode	j:PersonSexCode	ncic:SEXCodeType	A gender or sex of a person.	

JSON Name	NIEM Data Element	Type	Description	Comment
sexualOrientation	nc:PersonSexualOrientationText	nc:TextType	A target gender of the sexual interest of a person.	
skinToneCode	j:PersonSkinToneCode	ncic:SKINCodeType	A color or tone of the skin of a person.	
ssnIdentification	nc:PersonSSNIdentification	nc:IdentificationType	A unique identification reference to a living person; assigned by a country.	In the US and Canada, this is a 9-digit numeric identifier. e.g. Social Security Number, Social Insurance Number
weight	nc:PersonWeightMeasure	nc:WeightMeasureType	A measurement of the weight of a person.	
nationalityIso3166Alpha2Code	nc:PersonNationalityISO3166Alpha2Code	iso_3166:CountryAlpha2CodeType	A country of a person's citizenship or a country in which a person is deemed a national.	
accent	nc:PersonAccentText	nc:TextType	A manner of pronunciation; a way of pronouncing words that may indicate the place of origin or social background of the speaker.	
medicalCondition	nc:PersonMedicalCondition	nc:MedicalConditionType	A state of health for a person, ongoing or present.	
medicalDescription	nc:PersonMedicalDescriptionText	nc:TextType	A description of the overall health of a person.	

JSON Name	NIEM Data Element	Type	Description	Comment
medicationRequired	nc:PersonMedicationRequiredText	nc:TextType	A medication and dosage required for a person.	
mood	nc:PersonMoodDescriptionText	nc:TextType	A description of a state of feeling of a person.	e.g., temper, humor
physicalDisability	nc:PersonPhysicalDisabilityText	nc:TextType	A physical disability of a person.	
emergencyContact	nc:PersonEmergencyContactInformation	nc:ContactInformationType	A means of contacting someone in the event of an emergency.	

1153 ### 8.1.2 Vehicle Type

1154 **Table 8-2 Vehicle Type**

JSON Name	Data Element	Type	Description	Comment
condition	nc:ItemConditionText	nc:TextType	A state or appearance of an item.	e.g., new, used, damaged
description	nc:ItemDescriptionText	nc:TextType	A description of an item.	
owner	nc:ItemOwner	nc:EntityType	An entity which owns a property item.	Property owner
ownerAppliedID	nc:ItemOwnerAppliedID	niem-xs:string	An identifier applied to an item by the owner.	A number on a car to identify it within a fleet of cars.
serialIdentification	nc:ItemSerialIdentification	nc:IdentificationType	An identification inscribed on or attached to a part, collection of parts, or complete unit by the manufacturer.	
category	nc:ItemCategoryText	nc:TextType	A kind of property item.	

JSON Name	Data Element	Type	Description	Comment
colorPrimaryCode	j:ConveyanceColorPrimaryCode	ncic:VCOCodeType	A single, uppermost, frontmost, or majority color of a vehicle.	
colorSecondaryCode	j:ConveyanceColorSecondaryCode	ncic:VCOCodeType	A lowermost or rearmost color of a two-tone vehicle or a lesser color of a multicolored vehicle.	
image	nc:ItemImage	nc:ImageType	A binary representation of an image of an item.	
makeName	nc:ItemMakeName	nc:ProperNameTextType	A name of the manufacturer that produced an item.	
modelName	nc:ItemModelName	nc:ProperNameTextType	A name of a specific design or kind of item made by a manufacturer.	
modelYear	nc:ItemModelYearDate	niem-xs:gYear	A year in which an item was manufactured or produced.	
styleCode	j:VehicleStyleCode	ncic:VSTCodeType	A style of a vehicle.	
registrationPlateIdentification	j:ConveyanceRegistrationPlateIdentification	nc:IdentificationType	An identification on a metal plate fixed to a conveyance.	
isMotorized	nc:ConveyanceMotorizedIndicator	niem-xs:boolean	True if a conveyance is powered by a motor; false otherwise.	
isTowed	nc:ConveyanceTowedIndicator	niem-xs:boolean	True if a conveyance is towed; false otherwise.	
useAnsiD20Code	j:VehicleUseANSID20Code	aamva_d20:VehicleUseCodeType	A manner or way in which a vehicle is used.	
isCMV	nc:VehicleCMVIndicator	niem-xs:boolean	True if a vehicle is a commercial motor vehicle; false otherwise.	

JSON Name	Data Element	Type	Description	Comment
doorQuantity	nc:VehicleDoorQuantity	niem-xs:nonNegativeInteger	A number of doors on a vehicle.	
identification	nc:VehicleIdentification	nc:IdentificationType	A unique identification for a specific vehicle.	e.g., VIN, VIN number, registration, plate
makeCode	j:VehicleMakeCode	ncic:VMACodeType	A manufacturer of a vehicle.	
modelCode	j:VehicleModelCode	ncic:VMOCodeType	A specific design or class of vehicle made by a manufacturer.	
isRental	nc:ItemRentalIndicator	niem-xs:boolean	True if an item is rented; false otherwise.	Applies to short term rental service only.
towerAssociation	nc:VehicleTowerAssociation	nc:ItemEntityAssociationType	An association between a vehicle and the entity that towed it.	
vina	j:VehicleVINAText	nc:TextType	A Vehicle Identification Number Analysis; a combination of a vehicle make and model information.	
isWanted	j:VehicleWantedIndicator	niem-xs:boolean	True if a vehicle is being searched for by law enforcement; false otherwise.	

1155 ### 8.1.3 Date Type

1156 **Table 8-3 Date Type**

JSON Name	Data Element	Type	Description	Comment
date	nc:Date	niem-xs:date	A full date.	

JSON Name	Data Element	Type	Description	Comment
dateTime	nc:DateTime	niem-xs:dateTime	A full date and time.	
fiscalYear	nc:FiscalYearDate	nc:FiscalYearDateType	A year of a twelve month period that does not necessarily correspond to the calendar year.	
yearMonth	nc:YearMonthDate	niem-xs:gYearMonth	A year and month.	
marginOfError	nc:DateMarginOfErrorDuration	niem-xs:duration	A subjective assessment of the uncertainty of an estimated point by bounding an elements value with an estimated margin of error.	

1157 ### 8.1.4 Location Type

1158 **Table 8-4 Location Type**

JSON Name	Data Element	Type	Description	Comment
address	nc:Address	nc:AddressType	A postal location to which paper mail can be directed.	
addressGrid	nc:AddressGrid	nc:AddressGridType	A location identified by a unit of a grid system overlaid on an area.	
area	nc:LocationArea	nc:AreaType	A location identified by geographic boundaries.	
wgs84LocationCylinder	mo:WGS84LocationCylinder	mo:WGS84LocationCylinderType	A location identified by a cylinder oriented vertically and centered on a point described with WGS84 coordinates. If it is appropriate for the radius and	

JSON Name	Data Element	Type	Description	Comment
			half-height properties to represent an error value (for example, because the event is a laser-designated target), then the true event location follows a normal distribution such that the cylinder defines the one-sigma (p≈0.67)(almost equal to) deviation. (A cylinder with twice the volume would be the two-sigma (p≈0.95)(almost equal to) deviation, etc.) Otherwise, the cylinder encloses the full physical extent of the event.	
categoryCodeText	it:LocationCategoryCodeText	nc:TextType	A category of operation performed at a given location.	
categoryText	nc:LocationCategoryText	nc:TextType	A kind or functional description of a location.	
contact	nc:LocationContactInformation	nc:ContactInformationType	A set of contact information for a location.	
addressCrossStreet	nc:AddressCrossStreet	nc:CrossStreetType	A location identified by two or more streets which intersect.	
description	nc:LocationDescriptionText	nc:TextType	A description of a location.	
locationLandmark	nc:LocationLandmarkText	nc:TextType	A distinguishing physical feature at a location.	
name	nc:LocationName	nc:ProperNameTextType	A name of a location.	
floorNumber	nc:LocationFloorNumberText	nc:TextType	A floor number of a location.	

JSON Name	Data Element	Type	Description	Comment
uncertainty	mo:LocationUncertaintyArea	nc:AreaType	An area of uncertainty about a location.	

8.1.5 Image Type

Table 8-5 Image Type

JSON Name	Data Element	Type	Description	Comment
id	nc:BinaryID	niem-xs:string	An identifier that references a binary object.	
isAvailable	nc:BinaryAvailableIndicator	niem-xs:boolean	True if a binary is available; false if it is not.	
captureDate	nc:BinaryCaptureDate	nc:DateType	A date on which a binary object is captured or created.	This is the date an image was taken, an audio file was recorded, etc.
description	nc:BinaryDescriptionText	nc:TextType	A description of a binary object.	
format	nc:BinaryFormatText	nc:TextType	A file format or content type of a binary object.	
uri	nc:BinaryURI	niem-xs:anyURI	A URL or file reference of a binary object.	
size	nc:BinarySizeValue	nc:NonNegativeDecimalType	A size of a binary object in kilobytes.	
categoryText	nc:BinaryCategoryText	nc:TextType	A kind of object that has been encoded.	e.g., mug shot, driver license

JSON Name	Data Element	Type	Description	Comment
				picture, audio confession
height	nc:ImageHeightValue	niem-xs:nonNegativeInteger	A height of an image in pixels.	
width	nc:ImageWidthValue	niem-xs:nonNegativeInteger	A width of an image in pixels.	

1161 ### 8.1.6 LengthMeasure Type

1162 **Table 8-6 LengthMeasure Type**

JSON Name	Data Element	Type	Description	Comment
doubleValue	nc:MeasureDoubleValue	niem-xs:double	A double measurement value.	
doubleRange	nc:MeasureDoubleRange	nc:DoubleRangeType	A double measurement range.	
textList	nc:MeasureTextList	nc:StringListType	A list of measurement values, all using the same measurement method/device and of the same units.	
unitText	nc:MeasureUnitText	nc:TextType	A unit that qualifies the measurement value.	
method	nc:MeasureMethodText	nc:TextType	A method used to make a measurement.	
unitCode	nc:LengthUnitCode	unece:LengthCodeType	A unit of measure of a length value.	

8.1.7 Injury Type

Table 8-7 Injury Type

JSON Name	Data Element	Type	Description	Comment
categoryCode	j:InjuryCategoryCode	nc:TextType	A kind of bodily harm or injury.	
date	nc:InjuryDate	nc:DateType	A date on which an injury occurred.	In EIDO, InjuryDate should include DateRange/EndDate and DateRange/StartDate
description	nc:InjuryDescriptionText	nc:TextType	A description of an injury.	
locationCode	j:InjuryLocationNDEXCode	nc:TextType	A place on the body of a person where an injury occurred.	
severity	nc:InjurySeverityText	nc:TextType	A degree of the seriousness or intensity of an injury.	

8.1.8 Organization Type

Table 8-8 Organization Type

JSON Name	Data Element	Type	Description	Comment
activity	nc:OrganizationActivityText	nc:TextType	An activity that an organization is known or thought to be involved with.	e.g., law enforcement, supervision
branchName	nc:OrganizationBranchName	nc:TextType	A name of the chapter or branch by which an organization is	

JSON Name	Data Element	Type	Description	Comment
			known within a larger group of organizations.	
dayContact	nc:OrganizationDayContactInformation	nc:ContactInformationType	A means of contacting an organization during daytime hours.	
doingBusinessAsName	nc:OrganizationDoingBusinessAsName	nc:TextType	A name an organization uses for conducting business.	
emergencyContact	nc:OrganizationEmergencyContactInformation	nc:ContactInformationType	A means of contacting an organization in the event of an emergency.	
establishedDate	nc:OrganizationEstablishedDate	nc:DateType	A date an organization was started.	
eveningContact	nc:OrganizationEveningContactInformation	nc:ContactInformationType	A means of contacting an organization during evening or early night hours.	
identification	nc:OrganizationIdentification	nc:IdentificationType	An identification that references an organization.	For a school, this would be a school identifier; for a lien holder, this would be a lien holder identifier; for a court, this would be a court identifier.
localIdentification	nc:OrganizationLocalIdentification	nc:IdentificationType	An identification assigned at a local level to an organization.	

JSON Name	Data Element	Type	Description	Comment
location	nc:OrganizationLocation	nc:LocationType	A location of an organization.	
name	nc:OrganizationName	nc:TextType	A name of an organization.	
nightContact	nc:OrganizationNightContactInformation	nc:ContactInformationType	A means of contacting an organization during late-night hours.	
otherIdentification	nc:OrganizationOtherIdentification	nc:IdentificationType	An alternate identification assigned to an organization.	
primaryContact	nc:OrganizationPrimaryContactInformation	nc:ContactInformationType	A preferred means of contacting an organization.	
principalOfficial	nc:OrganizationPrincipalOfficial	nc:PersonType	A chief or high ranking executive of an organization.	
status	nc:OrganizationStatus	nc:StatusType	A status of an organization.	e.g., active, inactive
subUnit	nc:OrganizationSubUnit	nc:OrganizationType	A division of an organization.	e.g., department, group
taxIdentification	nc:OrganizationTaxIdentification	nc:IdentificationType	A tax identification assigned to an organization.	e.g., Federal Employer Identification Number, FEIN, an Employer Identification Number, EIN
abbreviation	nc:OrganizationAbbreviationText	nc:TextType	An abbreviation, acronym, or code for an organization name.	e.g., FBI, NCIC
description	nc:OrganizationDescriptionText	nc:TextType	A description of an organization.	

8.1.9 Identification Type

Table 8-9 Identification Type

JSON Name	Data Element	Type	Description	Comment
id	nc:IdentificationID	niem-xs:string	An identifier.	
categoryCode	j:PersonIDCategoryCode	ncic:MNUCodeType	A kind of identifier assigned to a person.	
categoryDescription	nc:IdentificationCategoryDescriptionText	nc:TextType	A description of a kind of identification.	
effectiveDate	nc:IdentificationEffectiveDate	nc:DateType	A date an identification takes effect.	This may or may not be the issue date.
expirationDate	nc:IdentificationExpirationDate	nc:DateType	A date after which an identification is no longer valid.	
jurisdiction	nc:IdentificationJurisdiction	nc:JurisdictionType	An area, region, or unit where a unique identification is issued.	For EIDO only, nc:JurisdictionText in JurisdictionType must be supported.
source	nc:IdentificationSourceText	nc:TextType	A person, organization, or locale which issues an identification.	
status	nc:IdentificationStatus	nc:StatusType	A status of an identification.	e.g., valid, expired, revoked, suspended, replaced, duplicate, lost

8.1.10 Status Type

Table 8-10 Status Type

JSON Name	Data Element	Type	Description	Comment
status	nc:StatusText	nc:TextType	A status or condition of something or someone.	
date	nc:StatusDate	nc:DateType	A date a status was set, effected, or reported.	
description	nc:StatusDescriptionText	nc:TextType	A description of a status or condition of something or someone.	
issuerIdentification	nc:StatusIssuerIdentification	nc:IdentificationType	An identification of a person or organization which assigns a status.	

8.1.11 Address Type

Table 8-11 Address Type

JSON Name	Data Element	Type	Description	Comment
fullText	nc:AddressFullText	nc:TextType	A complete address.	Usually includes street, city, state, and zip code information. Can also include other mailing specifics such as PO Box, apartment

JSON Name	Data Element	Type	Description	Comment
				number, or suite number.
recipientName	nc:AddressRecipientName	nc:TextType	A name of a person, organization, or other recipient to whom physical mail may be sent.	
buildingName	nc:AddressBuildingName	nc:TextType	A name of a specific building at an address to distinguish it from other buildings at the same site.	
deliveryPointId	nc:AddressDeliveryPointID	niem-xs:string	An identifier of a single place or unit at which mail is delivered.	
privateMailbox	nc:AddressPrivateMailboxText	nc:TextType	A private mailbox within a company.	e.g., mail stop code
secondaryUnit	nc:AddressSecondaryUnitText	nc:TextType	A particular unit within a larger unit or grouping at a location.	e.g., apartment number, suite number
routeName	nc:LocationRouteName	nc:TextType	A name and number of a postal route.	
street	nc:LocationStreet	nc:StreetType	A road, thoroughfare, or highway.	
city	nc:LocationCityName	nc:ProperNameTextType	A name of a city or town.	
countyCode	nc:LocationCountyCode	census:USCountyCodeType	A county, parish, vicinage, or other such geopolitical subdivision of a state.	
state	nc:LocationState	nc:StateType	A state, commonwealth, province, or other such	Only nc:LocationState FIPS5-2Code is

JSON Name	Data Element	Type	Description	Comment
			geopolitical subdivision of a country.	required to be supported in StateType.
country	nc:LocationCountry	nc:CountryType	A country, territory, dependency, or other such geopolitical subdivision of a location.	Only nc:LocationCountryISO3166Alpha2Code is required to be supported in CountryType.

8.1.12 Area Type

Table 8-12 Area Type

JSON Name	Data Element	Type	Description	Comment
wgs84LocationEllipse	mo:WGS84LocationEllipse	mo:WGS84EllipseType	An area region described by an ellipse specified by a point, major axis, minor axis, and rotation, using WGS84, meters, and decimal degrees.	
wgs84LocationExternalPolygon	mo:WGS84LocationExternalPolygon	mo:WGS84ExternalPolygonType	An area region described by a polygon with no interior region, using WGS84 coordinates.	

1176 ### 8.1.13 ContactInformation Type

1177 **Table 8-13 ContactInformation Type**

JSON Name	Data Element	Type	Description	Comment
emailId	nc:ContactEmailID	niem-xs:string	An electronic mailing address by which a person or organization may be contacted.	
instantMessenger	nc:ContactInstantMessenger	nc:InstantMessengerType	A user account for an instant messaging program by which a person or organization may be contacted.	
mailingAddress	nc:ContactMailingAddress	nc:AddressType	A postal address by which a person or organization may be contacted.	
radio	nc:ContactRadio	nc:ContactRadioType	A method of contacting a person or organization by messages over a radio.	
telephoneNumber	nc:ContactTelephoneNumber	nc:TelephoneNumberType	A telephone number for a telecommunication device by which a person or organization may be contacted.	
entity	nc:ContactEntity	nc:EntityType	An entity that may be contacted by using the given contact information.	
entityDescription	nc:ContactEntityDescriptionText	nc:TextType	A description of the entity being contacted.	This could be a title or function of the person with the given

JSON Name	Data Element	Type	Description	Comment
				contact information.
informationDescription	nc:ContactInformationDescriptionText	nc:TextType	A description of the contact information.	
responder	nc:ContactResponder	nc:PersonType	A third party person who answers a call and connects or directs the caller to the intended person.	

1178 ### 8.1.14 Entity Type

1179 **Table 8-14 Entity Type**

JSON Name	Data Element	Type	Description	Comment
organization	nc:EntityOrganization	nc:OrganizationType	An organization capable of bearing legal rights and responsibilities.	
person	nc:EntityPerson	nc:PersonType	A person capable of bearing legal rights and responsibilities.	

1180 ### 8.1.15 CrossStreet Type

1181 **Table 8-15 CrossStreet Type**

JSON Name	Data Element	Type	Description	Comment
description	nc:CrossStreetDescriptionText	nc:TextType	A description of a street intersection.	

JSON Name	Data Element	Type	Description	Comment
relativeLocation	nc:CrossStreetRelativeLocation	nc:RelativeLocationType	A location of something relative to a street intersection.	

8.1.16 PhysicalFeature Type

Table 8-16 PhysicalFeature Type

JSON Name	Data Element	Type	Description	Comment
generalCategory	nc:PhysicalFeatureGeneralCategoryText	nc:TextType	A general kind of physical feature.	e.g., scar, mark, tattoo, missing limb
categoryCode	j:PhysicalFeatureCategoryCode	nc:TextType	A specific kind of physical feature.	
description	nc:PhysicalFeatureDescriptionText	nc:TextType	A description of a physical feature.	
image	nc:PhysicalFeatureImage	nc:ImageType	A digital image of a physical feature.	
location	nc:PhysicalFeatureLocationText	nc:TextType	A location on a person's body of a physical feature.	

8.1.17 MedicalCondition Type

Table 8-17 MedicalCondition Type

JSON Name	Data Element	Type	Description	Comment
conditionText	nc:MedicalConditionText	nc:TextType	A state of health, ongoing or present.	e.g., scar, mark, tattoo, missing limb

JSON Name	Data Element	Type	Description	Comment
cause	nc:MedicalConditionCauseText	nc:TextType	A trigger that can initiate the onset of a medical condition.	e.g., drug, food, allergen
description	nc:MedicalConditionDescriptionText	nc:TextType	A description of a medical condition.	
isPresent	nc:MedicalConditionPresentIndicator	niem-xs:boolean	True if a medical condition currently exists; false otherwise.	
severity	nc:MedicalConditionSeverityText	nc:TextType	A degree to which a medical condition is affecting a person.	
dateRange	nc:MedicalConditionDateRange	nc:DateRangeType	A date range for the start and end of a medical condition.	

8.1.18 CommercialVehicle Type

Table 8-18 CommercialVehicle Type

JSON Name	Data Element	Type	Description	Comment
commercialVehicleCurrentlyTargetedIndicator	j:CommercialVehicleCurrentlyTargetedIndicator	niem-xs:boolean	True if the Performance and Registration Information Systems Management (PRISM) Carrier Target/History Indicator reveals the carrier is currently Targeted for increased enforcement action record; false otherwise.	
commercialVehiclePRISMAdditionDate	j:CommercialVehiclePRISMAdditionDate	niem-xs:date	A date on which the vehicle was added to the	

JSON Name	Data Element	Type	Description	Comment
			Performance and Registration Information Systems Management (PRISM) Vehicle File for safety monitoring.	
commercialVehicleMotorCarrier	j:CommercialVehicleMotorCarrier	j:MotorCarrierType	An organization providing commercial motor vehicle transportation for compensation.	
commercialVehicleCargoBodyCategoryCode	j:CommercialVehicleCargoBodyCategoryCode	mmucc:VehicleCargoBodyCategoryCodeType	A kind of body for buses and trucks more than 10,000 lbs (4,536 kg) GVWR.	
commercialVehicleConfigurationCode	j:CommercialVehicleConfigurationCode	mmucc:CommercialVehicleConfigurationCodeType	An indication of the general configuration of commercial motor vehicle.	

1188 ### 8.1.19 MotorCCarrier Type

1189 **Table 8-19 MotorCarrier Type**

JSON Name	Data Element	Type	Description	Comment
motorCarrierDOTIdentification	j:MotorCarrierDOTIdentification	nc:IdentificationType	An identification assigned by a federal or state department of transportation Agency, to uniquely identify a given carrier.	The number is posted on the side of commercial vehicles.

JSON Name	Data Element	Type	Description	Comment
				Currently the numbers are 8 digits long, the old numbers were 7 digits long. The Identification Jurisdiction and Effective Date may be used to hold supporting information.
motorCarrierHazMatTransportationIndicator	j:MotorCarrierHazMatTransportationIndicator	niem-xs:boolean	True if a given carrier transports Hazardous Materials (HAZMAT); false otherwise.	

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8.2 Appendix B—Mapping between vCard and NIEM Person Type

Table 8-20 Mapping between vCard and NIEM Person Type

vCard Data Properties	NIEM Data Element
General Properties:	
BEGIN	No mapping
END	No mapping
SOURCE	No mapping
KIND	No mapping
XML	No mapping
Identification Properties:	
FN	nc:PersonFullName
N	nc:PersonSurName
	nc:PersonGivenName
	nc:PersonMiddleName
	nc:PersonNamePrefixText
	nc:PersonNameSuffixText
NICKNAME	im:PersonNickName
PHOTO	nc:PersonDigitalImage
BDAY	nc:PersonBirthDate
ANNIVERSARY	No mapping
GENDER	j:PersonSexCode
ADR	nc:AddressPrivateMailboxText
	nc:AddressSecondaryUnitText
	nc:LocationStreet
	nc:LocationCityName
	nc:LocationPostalCode
	nc:LocationCountry
TEL	nc:FullTelephoneNumber

vCard Data Properties	NIEM Data Element
EMAIL	nc:Email
IMPP	nc:ContactInstantMessenger
LANG	nc:PersonPrimaryLanguage
TZ	No mapping
GEO	geo:LocationGeospatialPoint
TITLE	em:JobTitleOrRoleName
ROLE	em:JobTitleOrRoleName
ORG	nc:OrganizationName
MEMBER	nc:OrganizationBranchName
	nc:OrganizationPrimaryContactInformation
RELATED	No mapping except for emergency contact: nc:PersonEmergencyContactInformation
CATEGORIES	nc:PersonDescriptionText
NOTE	nc:PersonDescriptionText
PRODID	No mapping
REV	No mapping
SOUND	No mapping
UID	nc:PersonOtherIdentification
CLIENTPIDMAP	No mapping
URL	nc:URLID
VERSION	No mapping
KEY	nc:ContactWebsiteURI
FBURL	nc:ContactWebsiteURI
CALADRURI	nc:ContactWebsiteURI
CALURI	nc:ContactWebsiteURI

8.3 Appendix C—OpenAPI Schema

The schemas defined in this document are described using OpenAPI V3 [3], with JSON-ld exchange schemas for NIEM conformance. Only the component section of this schema is relevant to the EIDO JSON object described in this document. The normative reference can be found in NENA git hub repository available at https://github.com/NENA911/EIDO-JSON.

8.4 Appendix D—Shared Communication Resource Data Components

8.4.1 PTT-P25 Object

The PTT-P25 Data Component describes a radio talk path, typically a P25 or analog talkgroup on a conventional or trunked radio system. However, it may be operating in direct (talk-around) mode or through a conventional radio repeater. The only required fields are channelId, WACN, systemId, and groupMode. The channelAlias field is required if known. All other fields are optional. The schema for this object is available at https://github.com/NENA911/SharedCommunicationResources/blob/main/PTT-P25.yaml.

Table 8-21 PTT-P25 Data Component

JSON Name	Use	Min	Max	Description
mode	Optional	0	1	The mode for the talkpath: "A" (analog), "D" (digital), or "M" (mixed).
channelAlias	Optional	0	1	The alias or common name associated with a talk path or radio channel. Required if known. For example, 'LTAC-1'.
channelId	Required	1	1	The P25 talkgroup ID for the talkpath or the channel identifier, if available. May be in hex or decimal format.
channelUplinkFrequency	Optional	0	1	The uplink frequency for the talkpath, if operating in direct or conventional mode.
channelDownlinkFrequency	Optional	0	1	The downlink frequency for the talkpath, if operating in direct or conventional mode.
rxCode	Optional	0	1	The Network Access Code (NAC) or Continuous Tone-Coded Squelch System (CSTCSS) code for the downlink talkpath, if available. May

JSON Name	Use	Min	Max	Description
				be presented in hex or decimal format.
txCode	Optional	0	1	The Network Access Code (NAC) or Continuous Tone-Coded Squelch System (CSTCSS) code for the uplink talkpath, if available. May be presented in hex or decimal format.
channelEncrypted	Optional	0	1	Indicates whether the talk path is encrypted.
systemAlias	Optional	0	1	The alias or common name for radio system. Typically the name for a radio system. For example, 'Allied Matrix for Emergency Response'.
WACN	Required	1	1	The Wide Area Communications Network code for the system. This, with the SystemId, uniquely identifies a trunked radio system.
systemId	Required	1	1	The system ID for a trunked radio system. This, with the WACN, uniquely identifies a trunked radio system.
units	Optional	0	*	Unit Data Component
talkMode	Optional	0	1	Whether the talk path is a group or private (to a designated unitId) call. Typically, talk paths are group calls.
groupMode	Required	1	1	The type of group resource if the talk path is a group resource. Allowed values are "Conventional" and "trunked".

8.4.1.1 Unit Data Component

Table 8-22 Unit Data Component

JSON Name	Use	Min	Max	Description
alias	Optional	0	1	The common name or alias for units assigned to the talk path for the

JSON Name	Use	Min	Max	Description
				incident. For example, 'Cruiser-2020-Mobile' or 'Dern Handy'.
terminalId	Optional	0	1	The unique ID for the radio terminal assigned to the talk path for the incident. Can be presented in hex or decimal.
unitId	Optional	0	1	The globally unique NG9-1-1 ID for individual units assigned to the talk path for the incident. This is the ID contained in the unit's PCA-traceable validated NG9-1-1 Certificate.

8.4.2 HTTP-URL Data Component

Generic Shared Communication resource that can be accessed via a URL using an HTTP enabled application.

Table 8-23 HTTP-URL Data Component

JSON Name	Use	Min	Max	Description
URL	Required	1	1	The URL that can be input into the HTTP enabled application to access the shared communication resource.

8.4.3 SIP-TEL-URI Data Component

SIP or TEL URI that can be used to reach a conference bridge.

Table 8-24 SIP-TEL-URI Data Component

JSON Name	Use	Min	Max	Description
URI	Required	1	1	The URI that can be used to reach the conference bridge.

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